



清华大学电子工程系
Department of Electronic Engineering, Tsinghua University

严谨 · 勤奋 · 求实 · 创新

数字系统设计

02-深度神经网络DNN

贾弘洋 & 刘勇攀

2025/02/25





- Artificial intelligence, machine learning, and deep learning
- Building a machine learning algorithm:
 - Dataset
 - Cost function
 - Optimization function
 - Model
- Deep learning to automatically extract hierarchical data
 - Better at handle high-dimensional data
 - Key differences are in the Model
 - Multiple layers, i.e., deep
 - Linear + non-linear layers

深度学习教程

- <https://www.deeplearningbook.org>

PyTorch教程

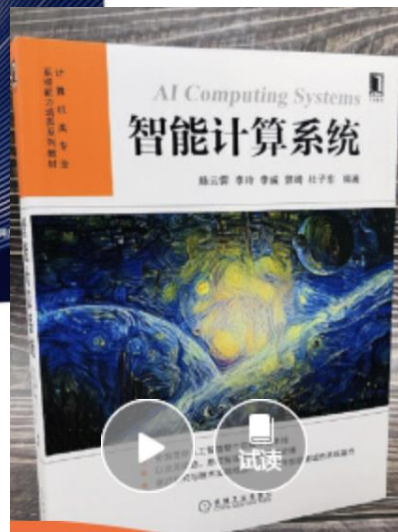
- <https://pytorch.org/tutorials/>

中文教材

- 昇腾AI处理器架构与编程

课程网站

- 论文阅读
- 硬件学习资源更新





- **智能算法与数据集导论**

- 智能模型
- 数据集
- 成本函数
- 优化算法
- 整体应用

- **人工智能模型导论**

- 智能模型评价
- 应用模型介绍

深度学习神经网络的训练和推理



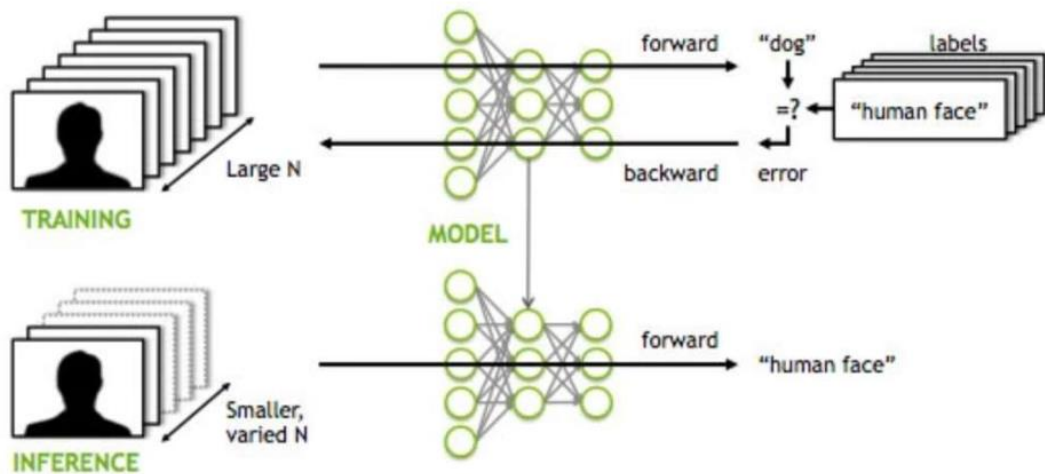
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- Training

- Dataset
- Cost function
- Optimization function
- Model

- Inference

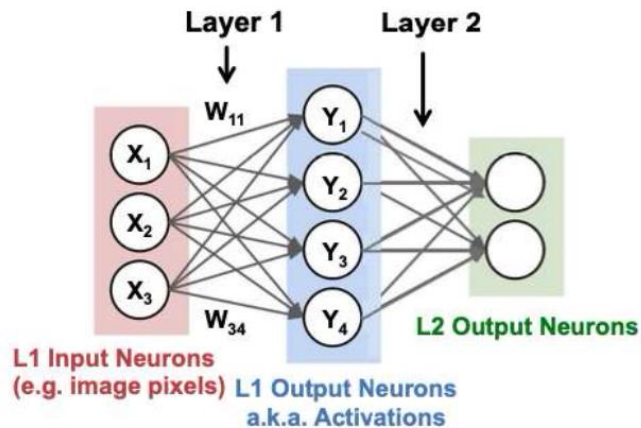
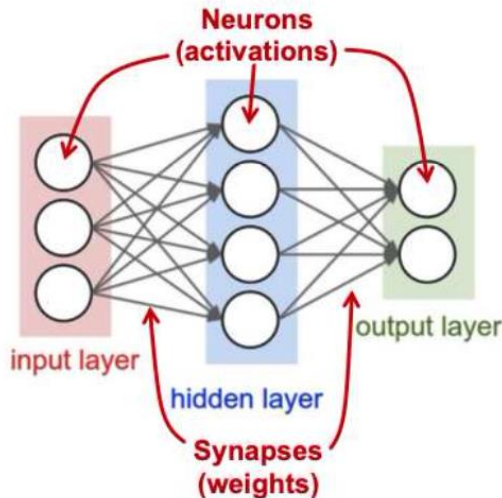
- Dataset
- Model

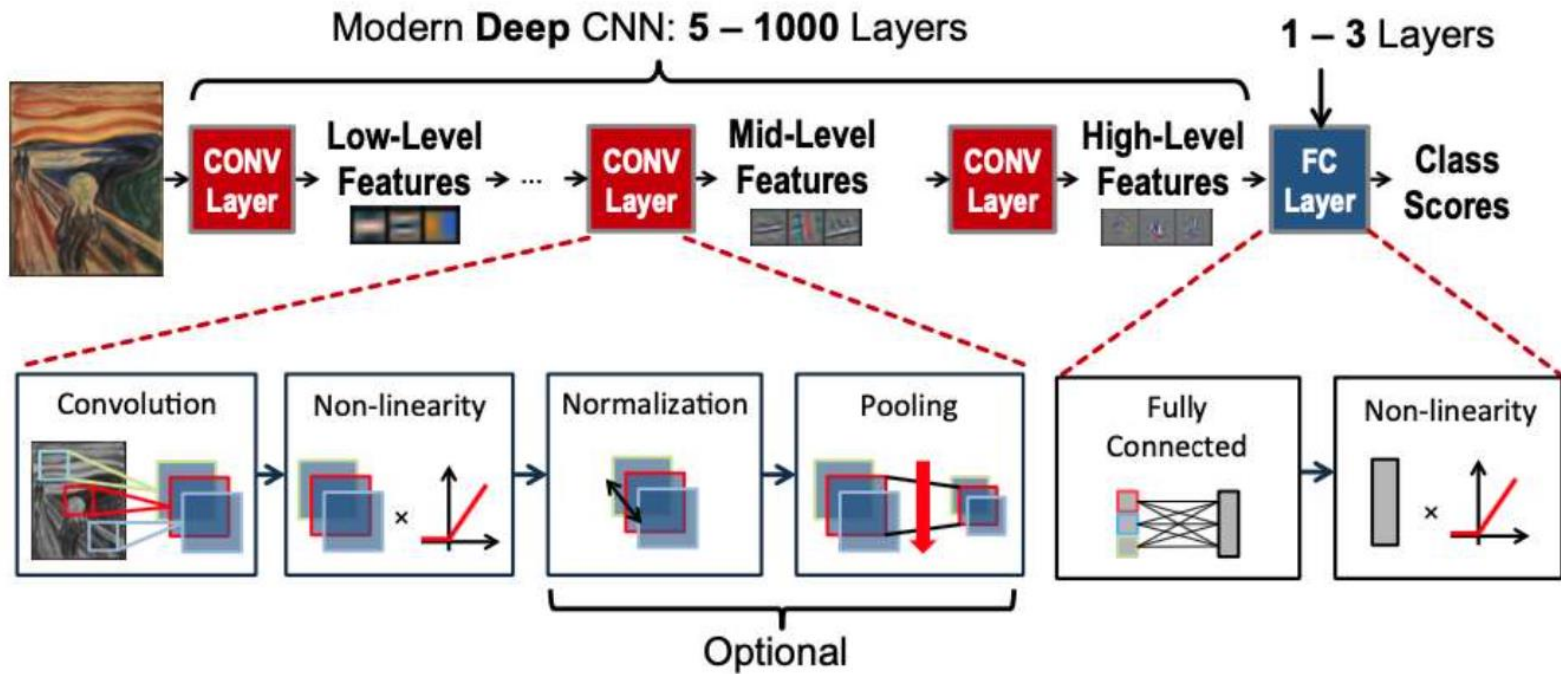




- Model
- Dataset
 - Problem dependent
 - Training set, validation set, test set
- Cost function
 - Convex function
 - Backpropagation (using chain rule)
- Optimization

- A subset of machine learning models
 - Deep -> multiple layers
 - Linear model + nonlinear transformation



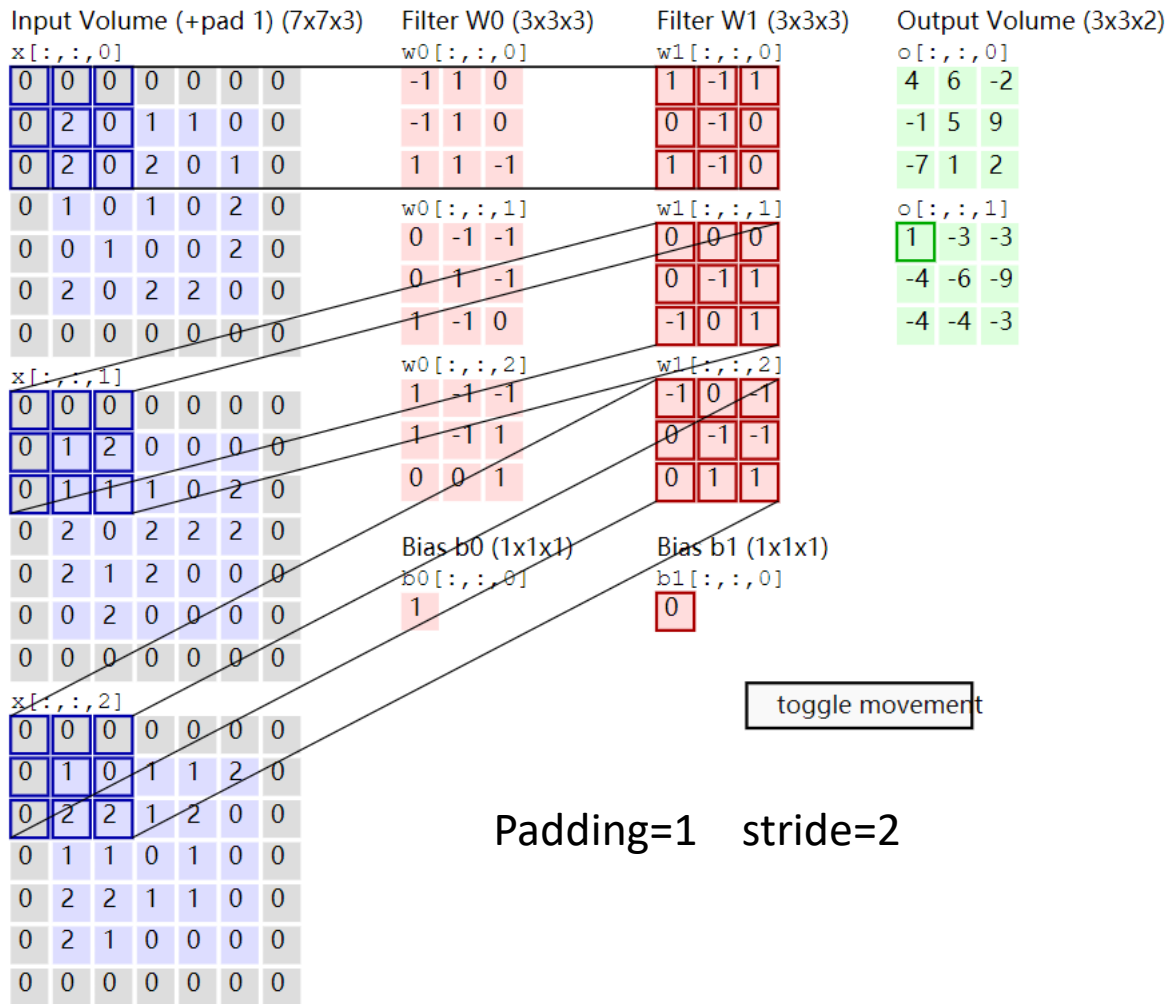


线性卷积层

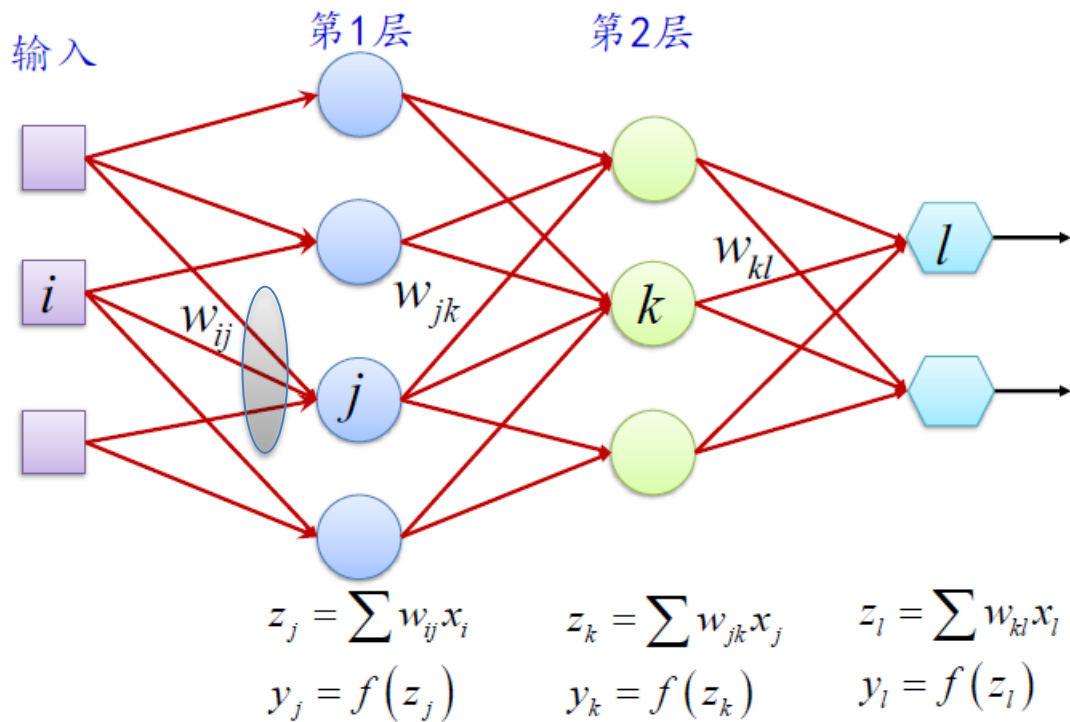
- ## ● 公式

$$H(x, y) = I * K(x, y) :$$

$$= \sum_m \sum_n I(x+m, y+n) K(m, m)$$



线性全连接层



非线性函数-激活函数



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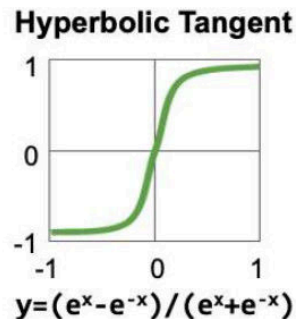
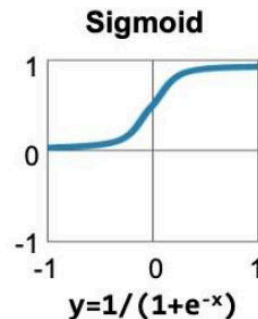
- Rectified Linear Units (ReLU)

- $y(x) = \max\{0, x\}$

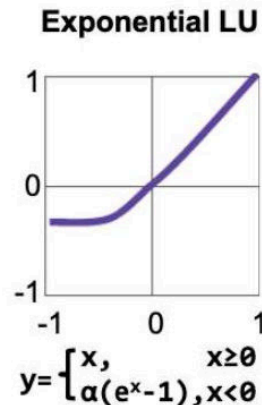
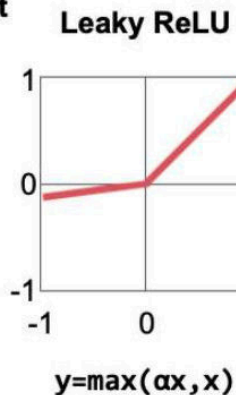
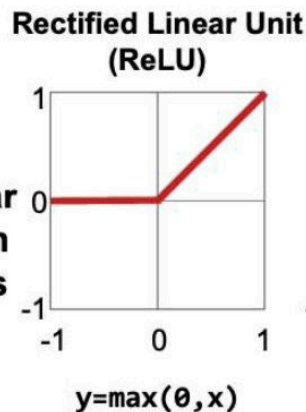
- Benefits:

- Reduce the likelihood of the gradient vanishing problem.
 - Adding more sparsity/regularization
 - Easy to compute

Traditional
Non-Linear
Activation
Functions



Modern
Non-Linear
Activation
Functions

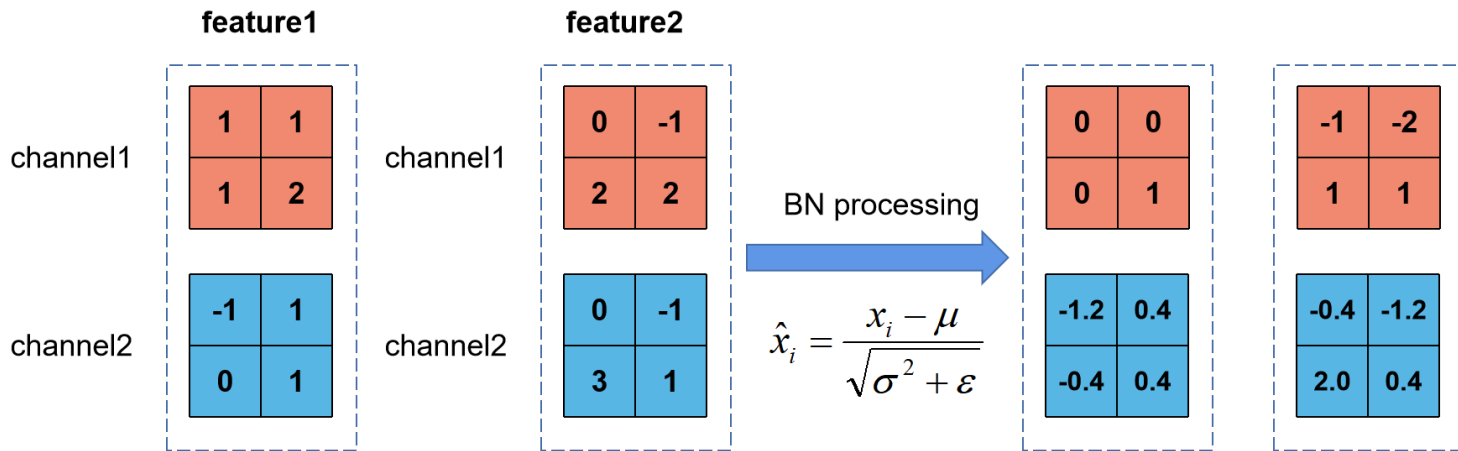


$\alpha = \text{small const. (e.g. 0.1)}$

非线性函数-正则化函数



- Batch normalization



$$x^{(1)} = \{1, 1, 1, 2, 0, -1, 2, 2\}$$

$$x^{(2)} = \{-1, 1, 0, 1, 0, -1, 3, 1\}$$

$$\mu_1 = \frac{1}{m} \sum_{i=1}^m x_i^{(1)} = 1$$

$$\mu_2 = \frac{1}{m} \sum_{i=1}^m x_i^{(2)} = 0.5$$

$$\sigma_1^2 = \frac{1}{m} \sum_{i=1}^m (x_i^{(1)} - \mu_1)^2 = 1$$

$$\sigma_2^2 = \frac{1}{m} \sum_{i=1}^m (x_i^{(2)} - \mu_2)^2 = 1.5$$



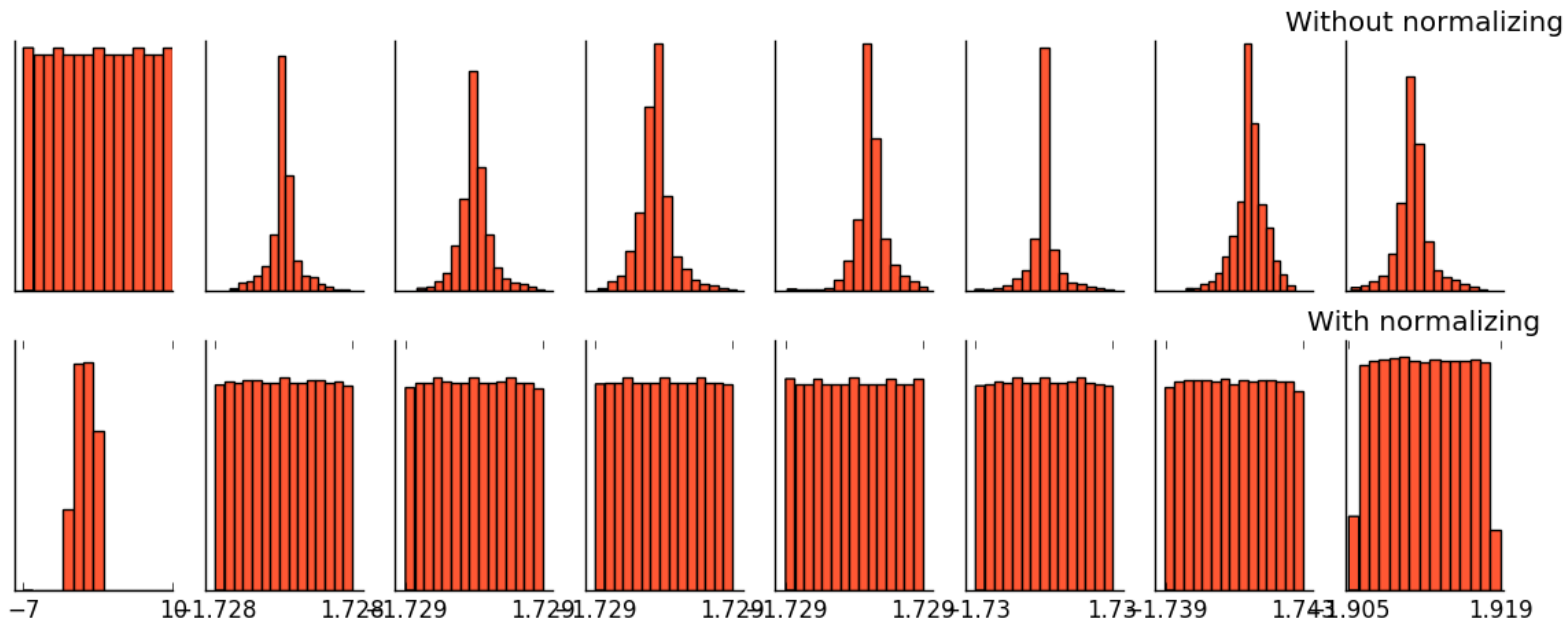
$$\mu = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \quad \sigma^2 = \begin{bmatrix} 1 \\ 1.5 \end{bmatrix}$$

非线性函数-正则化函数



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- Batch normalization有利于改善分布

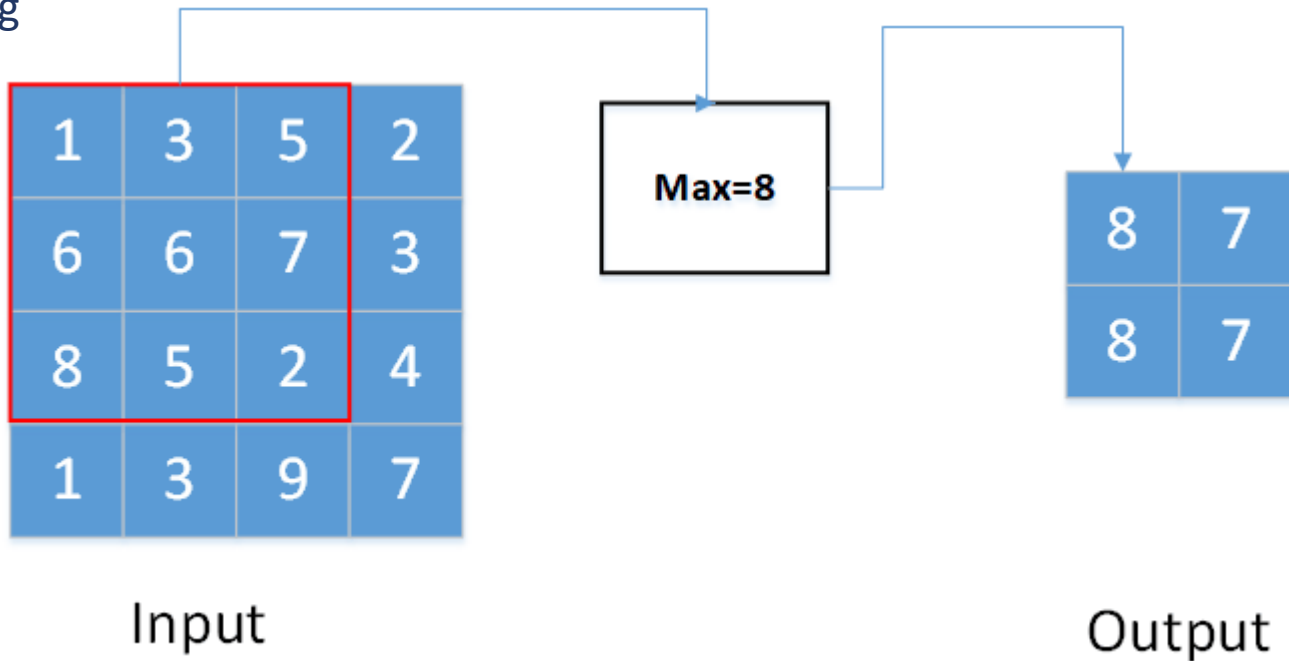


非线性函数-池化函数



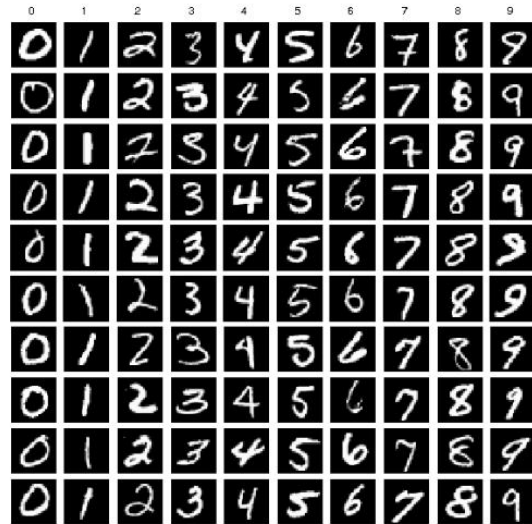
- 池化函数：只保留最主要的特征。

MaxPooling

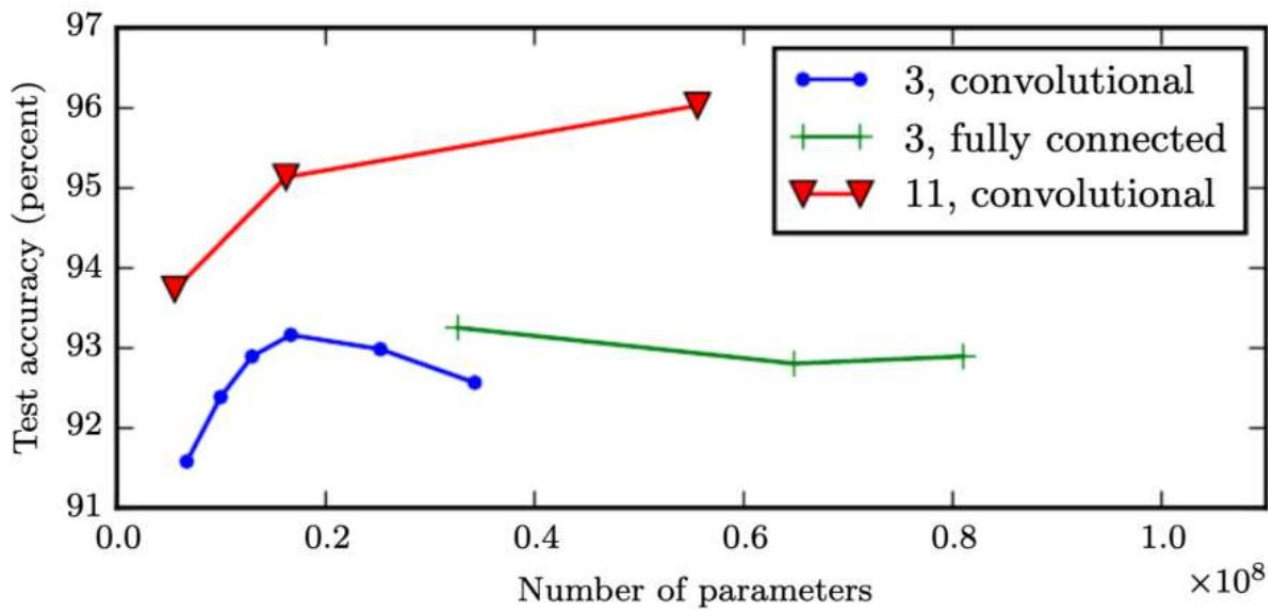


The diagram illustrates the architecture of a Convolutional Neural Network (CNN). It starts with an **Input** image of a handwritten letter 'K'. This input is processed through **Convolutions**, which generate a stack of **Image Maps** (feature maps). These maps are then processed through **Subsampling** (pooling) to reduce their dimensions. The resulting smaller feature maps are then passed through **Fully Connected** layers, which connect every neuron in one layer to every neuron in the next, leading to the final **Output**.

Illustration of LeCun et al. 1998 from CS231n 2017 Lecture 1



- Depth and width of DNNs



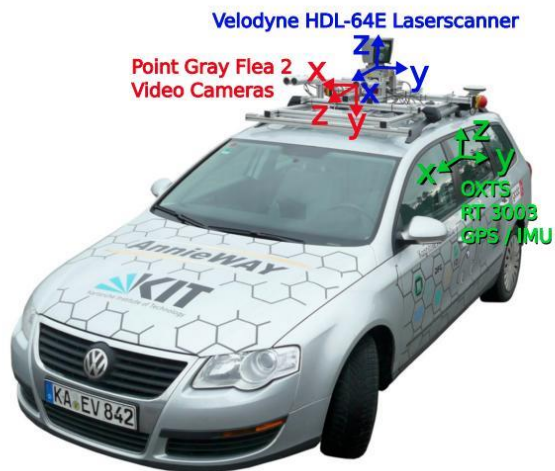


- Model
- Dataset
 - Problem dependent
 - Training set, validation set, test set
- Cost function
 - Convex function
 - Backpropagation (using chain rule)
- Optimization

/Dataset



Kitti
Dataset



Cityscape Dataset



Microsoft COCO

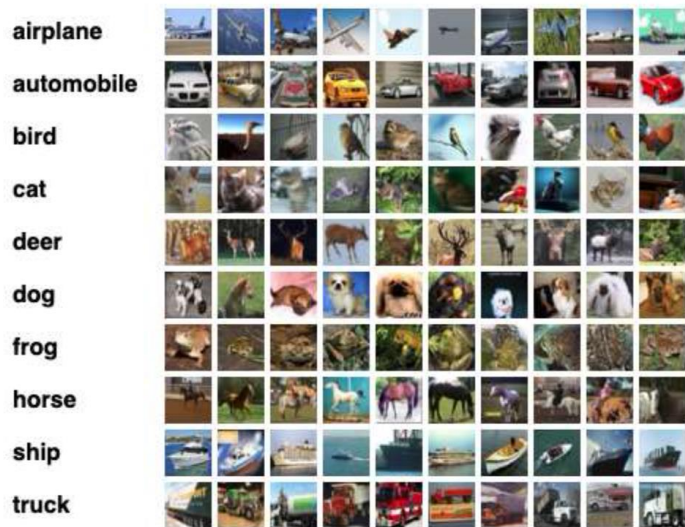


标准数据集：图像分类数据集



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CIFAR10/100（图像分类）



ImageNET（图像分类）

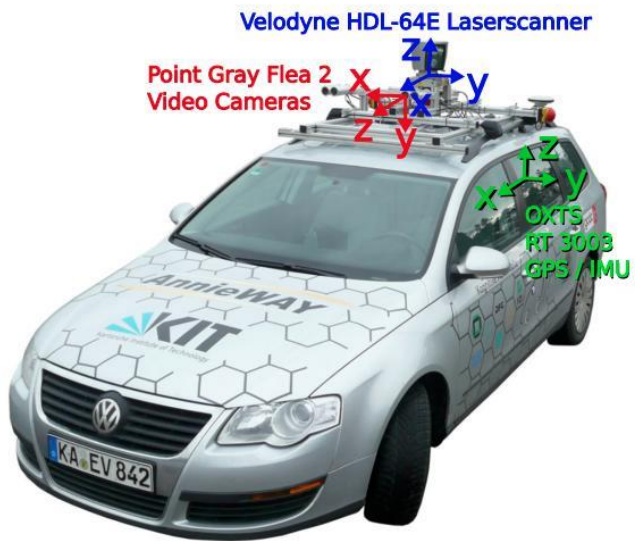


标准数据集：自动驾驶数据集



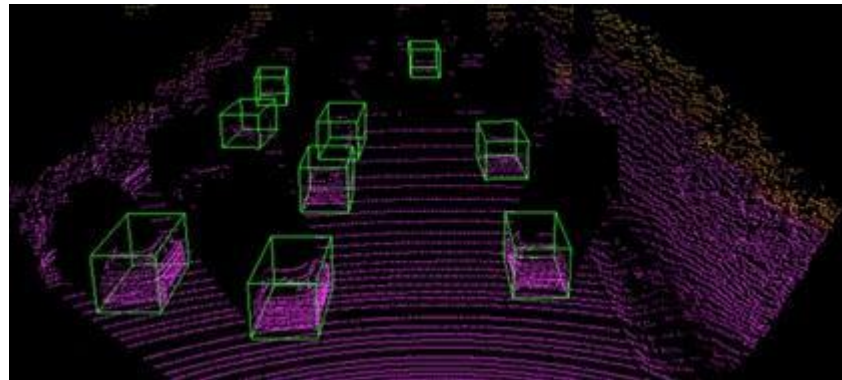
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Kitti/nuscenes数据集

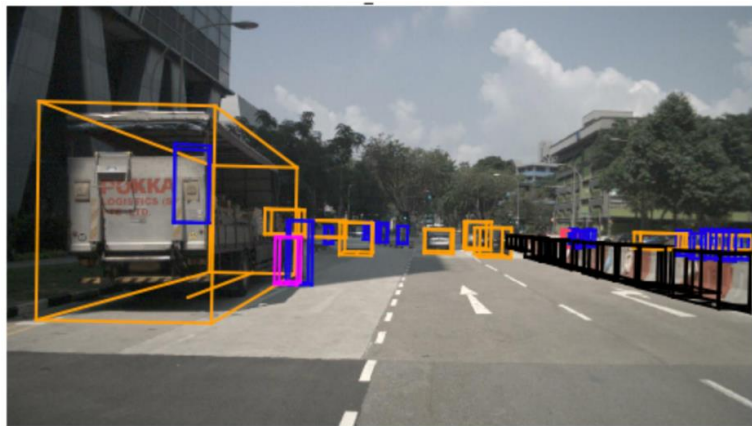


数据采集车

雷达数据



相机数据

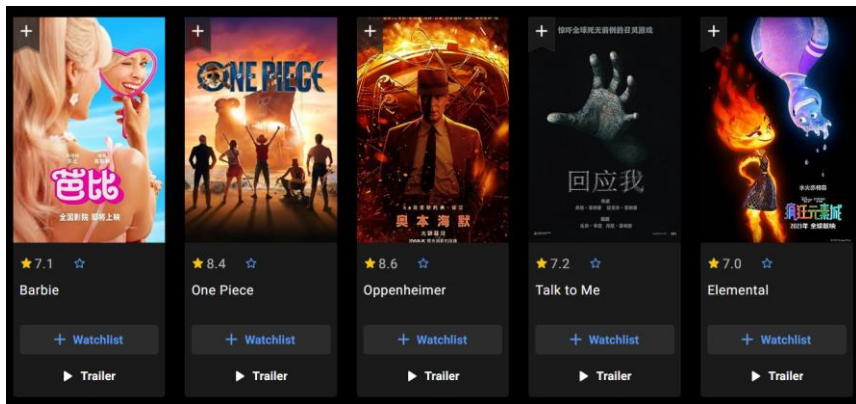


标准数据集：自然语言处理数据集

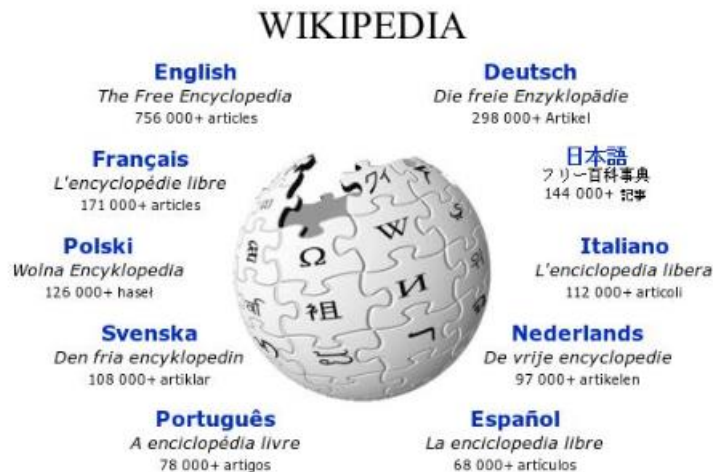


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IMDB（影评情感分析）



WikiText（语言模型训练）



数据集建立：自定义数据集（猫狗二分类）



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数据采集



数据标签

猫

猫

猫

狗

狗

狗

数据清洗



数据增强



数据分割





- Network
- Dataset
 - Problem dependent
 - Training set, validation set, test set
- **Cost function**
 - Convex function
 - Backpropagation (using chain rule)
- Optimization

/ Cost function

- 成本（损失）函数定义

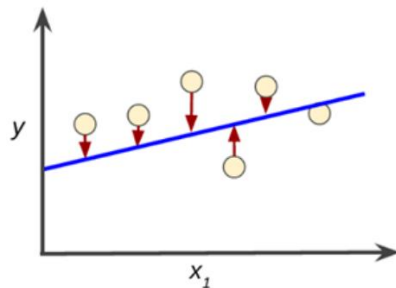
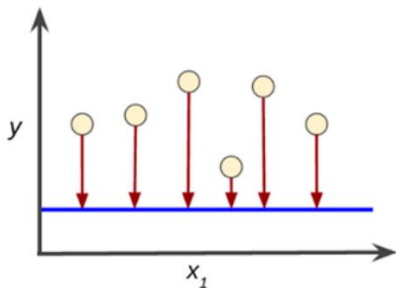
- 成本函数度量模型预测与真实值之间的误差

- 作用一：确定优化方向

- 作用二：度量模型好坏

- 成本函数属性

- 凸函数：全局最优=局部最优
- 可导：梯度最优
- 鲁棒性：异常值不敏感
- 光滑和单调
- 数据不变性：改变输入序列，不影响输出值



/ Cost function

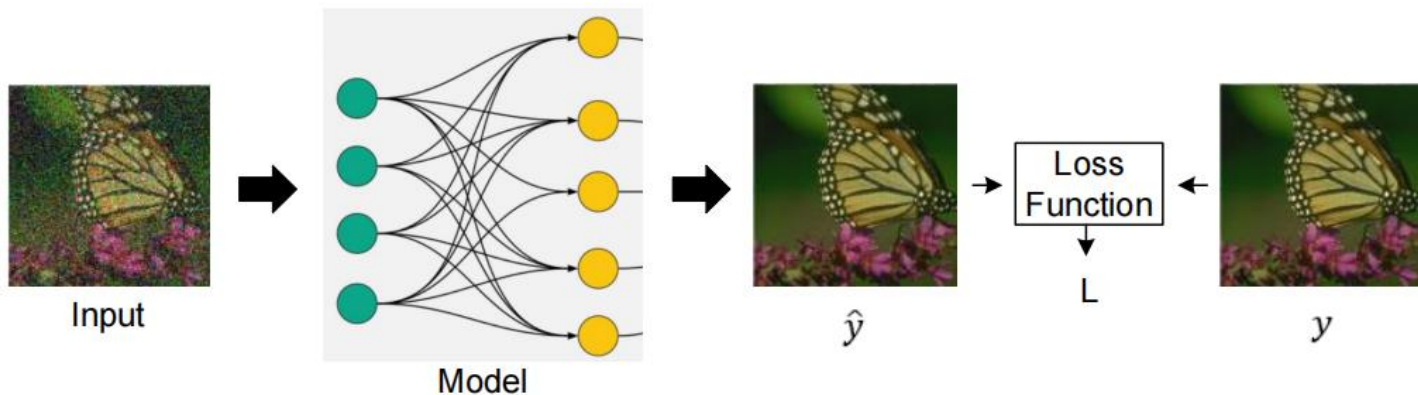
- 回归成本（损失）函数

- L-1 loss

$$MAE = \frac{1}{m} \sum_{i=1}^m |y_i - f(x_i)|$$

- MSE loss

$$MSE = \frac{1}{m} \sum_{i=1}^m (y_i - f(x_i))^2$$



/ Cost function



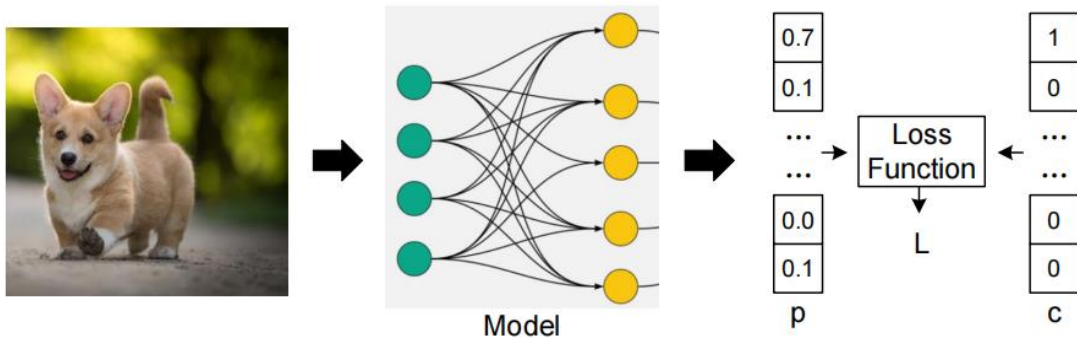
- 分类成本（损失）函数

- Cross Entropy loss

$$L = \frac{1}{N} \sum_i L_i = -\frac{1}{N} \sum_i \sum_{c=1}^M y_{ic} \log(p_{ic})$$

其中:

- M ——类别的数量
- y_{ic} ——符号函数（0 或 1），如果样本 i 的真实类别等于 c 取 1，否则取 0
- p_{ic} ——观测样本 i 属于类别 c 的预测概率





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梯度优化算法介绍



- 设计优化算法降低成本函数，即优化目标函数
- 梯度优化算法（步长 $\eta > 0$ ）

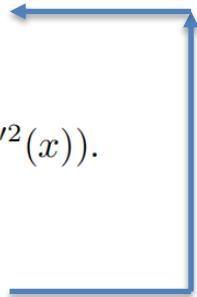
$$f(x + \epsilon) = f(x) + \epsilon f'(x) + \mathcal{O}(\epsilon^2).$$



$$f(x - \eta f'(x)) = f(x) - \eta f'^2(x) + \mathcal{O}(\eta^2 f'^2(x)).$$



$$x \leftarrow x - \eta f'(x)$$

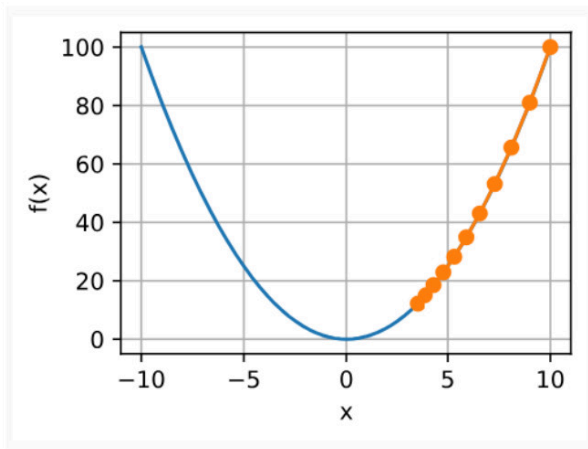


终止条件：梯度足够小或者迭代次数达到上限

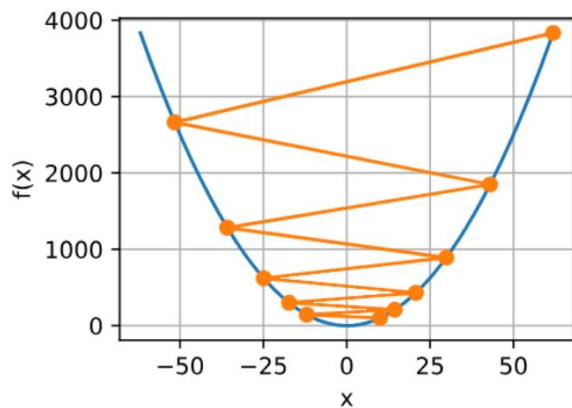
梯度优化算法的步长选择



- 设计优化算法降低成本函数，即优化目标函数
- 学习率 η 的选择



过小



过大

随机梯度优化算法



- SGD: 随机梯度优化降法
- $f(\mathbf{x})$: 目标函数
- $\mathbf{x}$: 输入向量
- N : 样本数 (来自于数据集)
- 随机梯度是对梯度良好估计
- 降低了训练复杂度

$$f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}).$$



$$\nabla f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\mathbf{x}).$$



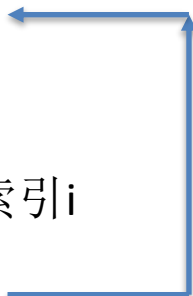
随机均匀采样一个索引 i



$$\mathbf{x} \leftarrow \mathbf{x} - \eta \nabla f_i(\mathbf{x}),$$



计算复杂度 $O(n) \rightarrow O(1)$

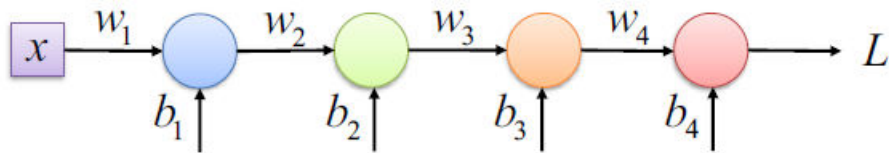


深度学习网络函数梯度优化



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- 链式法则求解梯度



圆圈表示sigmoid激活函数 $\sigma()$

$$\frac{\partial L}{\partial b_1} = \sigma'(b_1)w_2\sigma'(b_2)w_3\sigma'(b_3)w_4\frac{\partial L}{\partial b_4}$$

- 梯度下降更新参数

$$x \leftarrow x - \gamma \cdot \nabla$$

γ 被称为学习率

高级优化算法概述



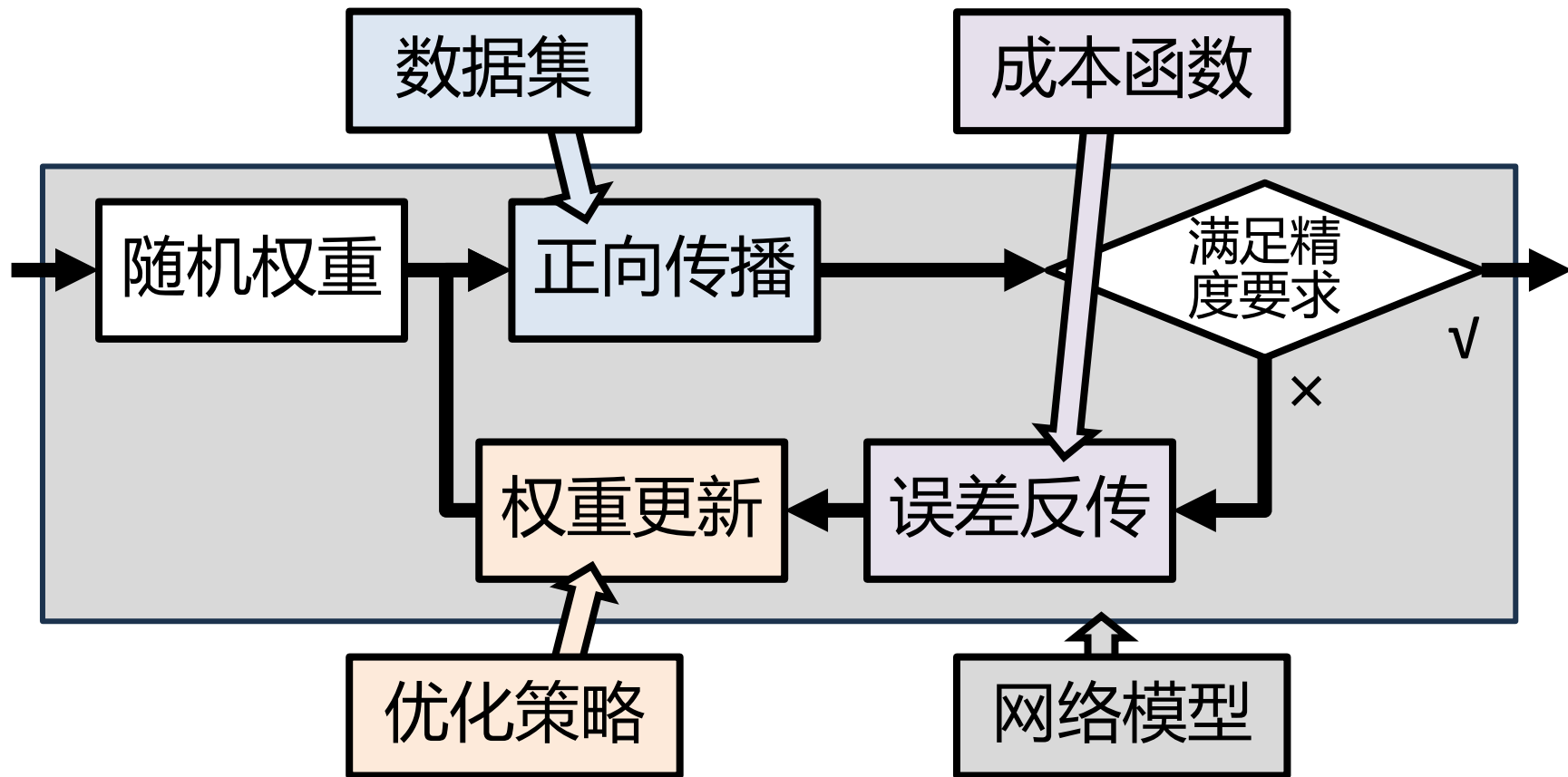
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- 动量法：计算过去的平均梯度
- AdaGrad法：累加梯度方差
- RMSProp：按时间降低学习率
- Adam算法：指数加权移动平均值算梯度动量和二次距
- 总结：SGD简单，但训练过程变长，自适应算法会更高效（Adam法）

智能网络训练流程



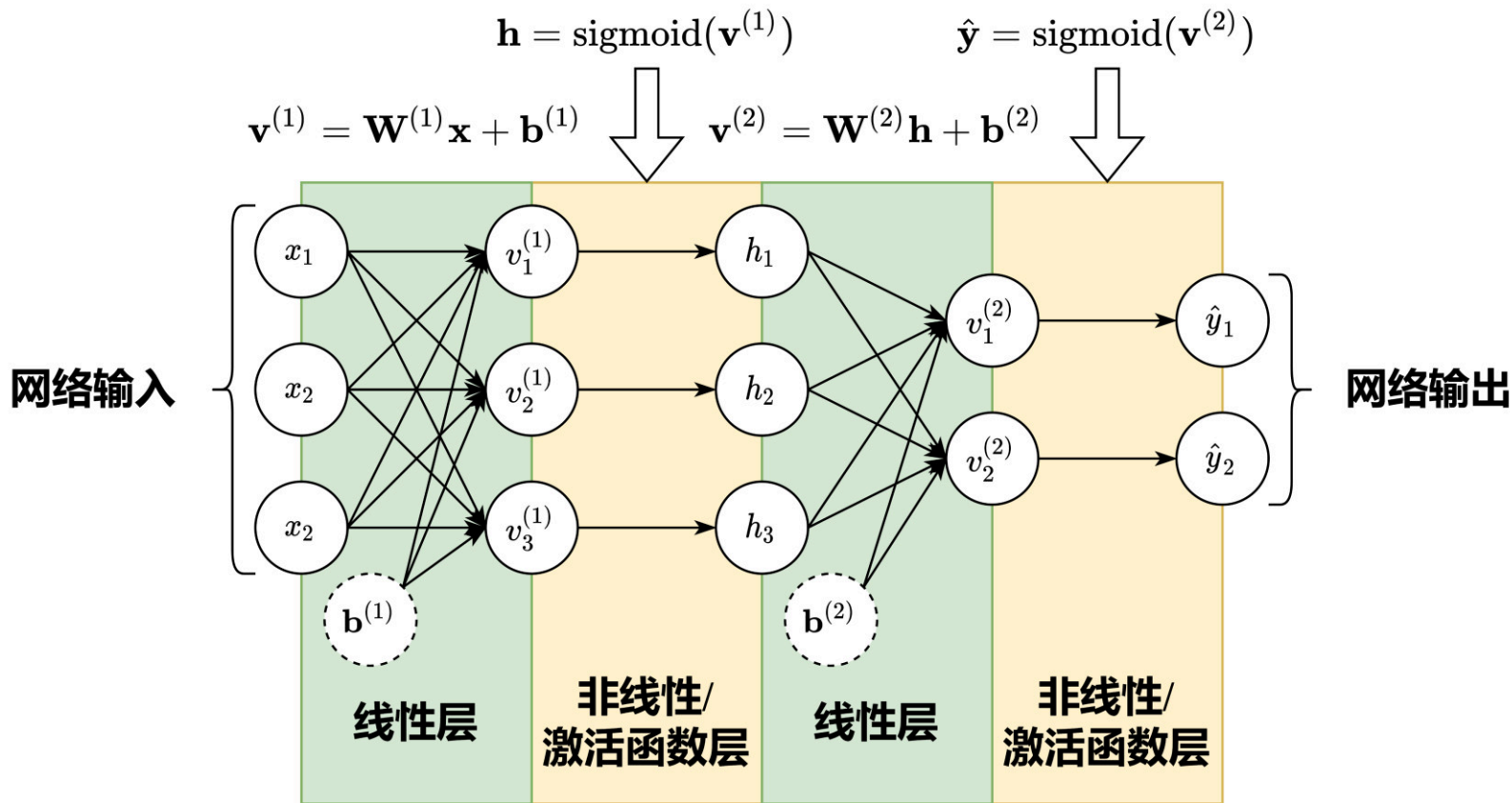
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智能网络训练流程：正向传播



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智能网络训练流程：误差反传



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以权重 $W_{2,1}^{(2)}$ 为例

传播方法：**链式法则**

$$\frac{\partial L}{\partial W_{2,1}^{(2)}} = \frac{\partial L}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial v_1^{(2)}} \cdot \frac{\partial v_1^{(2)}}{\partial W_{2,1}^{(2)}}$$

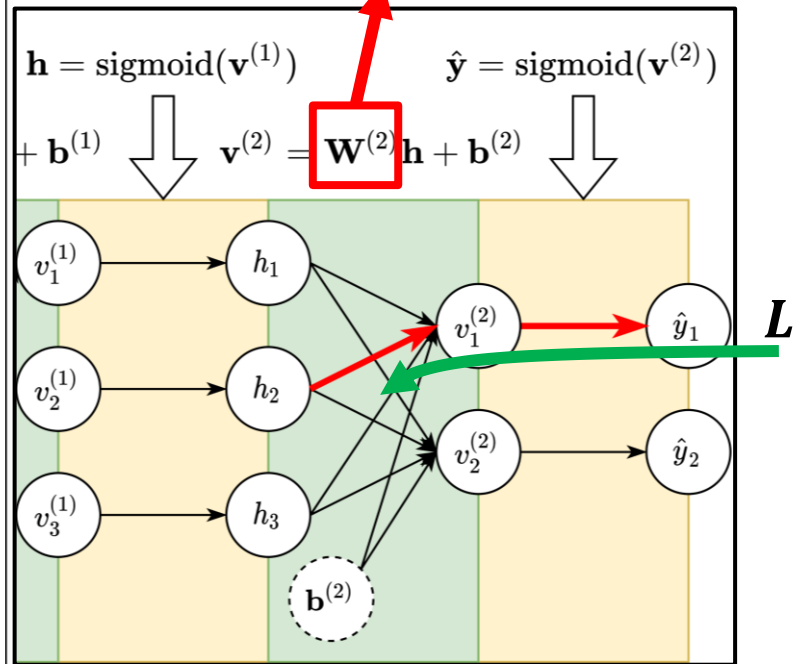
其中：

均方误差成本函数

$$L = \frac{1}{2} (y_1 - \hat{y}_1)^2 + \frac{1}{2} (y_2 - \hat{y}_2)^2 \Rightarrow \frac{\partial L}{\partial \hat{y}_1} = -(y_1 - \hat{y}_1)$$

$$\hat{y}_1 = \text{sigmoid}(v_1^{(2)}) \Rightarrow \frac{\partial \hat{y}_1}{\partial v_1^{(2)}} = \hat{y}_1(1 - \hat{y}_1)$$

$$v_1^{(2)} = W_{1,1}^{(2)} h_1 + W_{2,1}^{(2)} h_2 + W_{3,1}^{(2)} h_3 + b_1^{(2)} \Rightarrow \frac{\partial v_1^{(2)}}{\partial W_{2,1}^{(2)}} = h_2$$



→ 正向传播 → 误差反传

智能网络训练流程：权重更新



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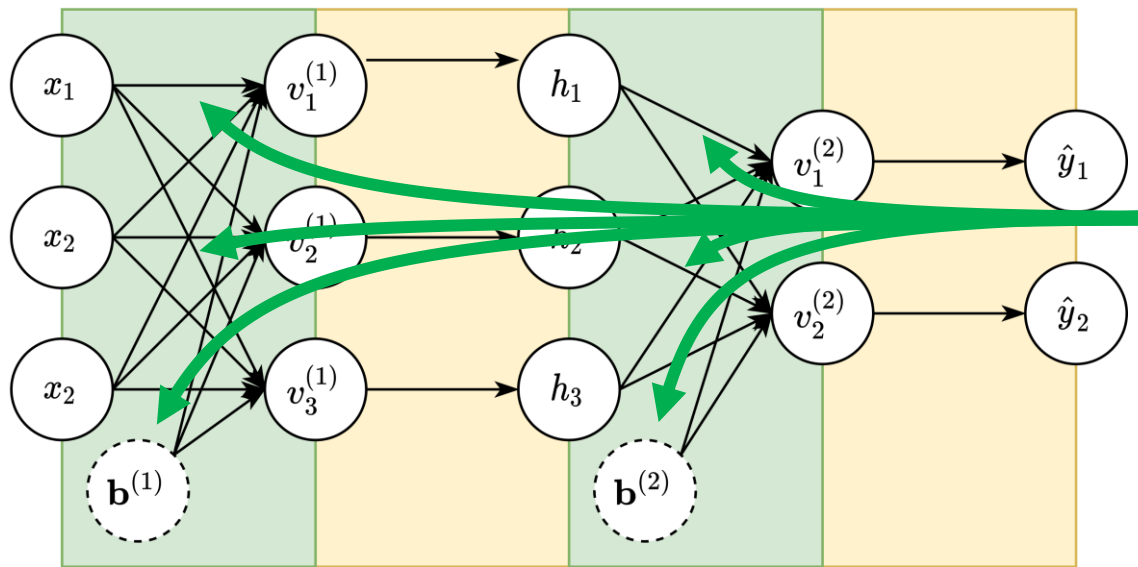
$$W_{i,j}^{(k)} \leftarrow W_{i,j}^{(k)} - \eta \frac{(\partial L)}{\partial W_{i,j}^{(k)}}, \quad b_i^{(k)} \leftarrow b_i^{(k)} - \eta \frac{(\partial L)}{\partial b_i^{(k)}}$$

$$\mathbf{v}^{(1)} = \mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)}$$

$$\mathbf{v}^{(2)} = \mathbf{W}^{(2)} \mathbf{h} + \mathbf{b}^{(2)}$$

$$\mathbf{h} = \text{sigmoid}(\mathbf{v}^{(1)})$$

$$\hat{\mathbf{y}} = \text{sigmoid}(\mathbf{v}^{(2)})$$



根据**链式法则**，求得
每一个权重的梯度：

$$\frac{\partial L}{\partial W_{i,j}^{(k)}}, \frac{\partial L}{\partial b_i^{(k)}}$$



- **智能算法与数据集导论**

- 智能模型
- 数据集
- 成本函数
- 优化算法
- 整体应用

- **人工智能模型导论**

- 智能模型评价
- 应用模型介绍

/评价指标



- 回归问题指标
 - PSNR/SSIM

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

- 分类问题指标
 - Top1 accuracy
 - Top5 accuracy

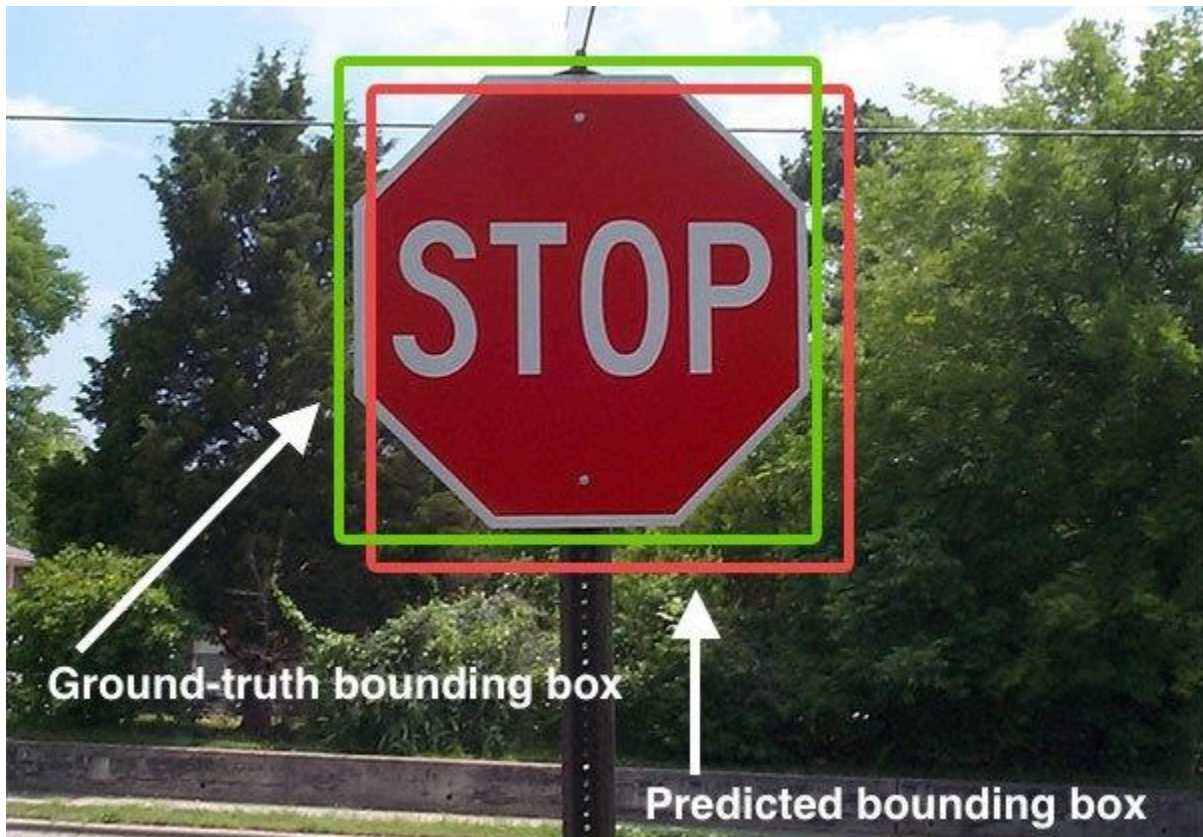


CAT

评价指标-IoU



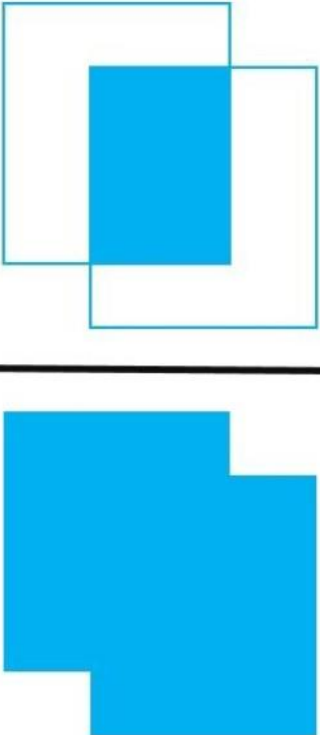
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/评价指标-IoU



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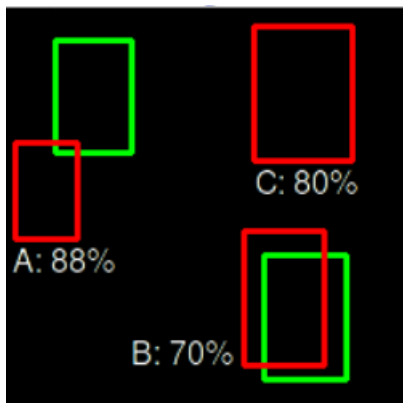
$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}$$


检测任务评价指标mAP



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- 在实际的检测任务中，可能不止有一个待检测物体，此时模型的输出是什么？该如何评价检测效果呢？
- 模型将输出多个候选框，并附带置信度。



如图所示，绿框为待检测物体的边框，红框为模型输出的检测边框与置信度。

显然，这里有漏检和错检。

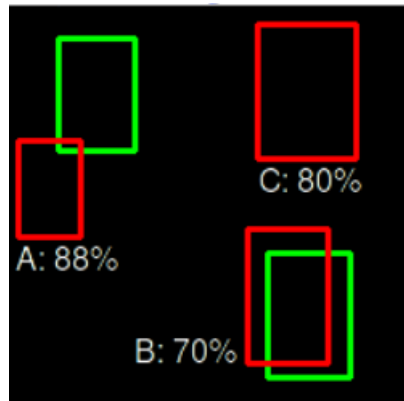
检测任务评价指标mAP

- mAP: mean average precision, 是进行多物体检测的通用指标, 其计算涉及召回率(或称查全率)及精度(或称查准率)的计算。
- TP: 与任一待检测绿框IoU>0.5的红框数量。检测为真且实际为真。
- FP: 与所有绿框IoU均<0.5的红框数量。检测为真但实际为假。
- FN: 与任一红框IoU均<0.5的绿框数量。检测为假但实际为真。

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{TP}{\text{all detections}}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{TP}{\text{all ground truths}}$$

测试结果	实际情况	
	正例(Positive)	反例(Negative)
正例(Positive)	真正例(True positive, TP)	假正例(False positive, FP)
反例(Negative)	假反例(False negative, FN)	真反例(True positive, TN)



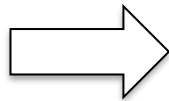
检测任务评价指标mAP



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- 假设模型对某一输入图像进行检测，输出了二十个检测框及各自的置信度。
- 我们可以根据IoU计算其是否检测到待检测物体，并按照置信度进行排序。

id	score	gt_label
1	0.23	0
2	0.76	1
3	0.01	0
4	0.91	1
5	0.13	0
6	0.45	0
7	0.12	1
8	0.03	0
9	0.38	1
10	0.11	0
11	0.03	0
12	0.09	0
13	0.65	0
14	0.07	0
15	0.12	0
16	0.24	1
17	0.1	0
18	0.23	0
19	0.46	0
20	0.08	1



id	score	gt_label
4	0.91	1
2	0.76	1
13	0.65	0
19	0.46	0
6	0.45	0
9	0.38	1
16	0.24	1
1	0.23	0
18	0.23	0
5	0.13	0
7	0.12	1
15	0.12	0
10	0.11	0
17	0.1	0
12	0.09	0
20	0.08	1
14	0.07	0
8	0.03	0
11	0.03	0
3	0.01	0

/ 检测任务评价指标mAP



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- 置信度衡量检测框是否被模型认为是有效的自信程度，可以作为滤除无效框的标准。

id	score	gt_label
4	0.91	1
2	0.76	1
13	0.65	0
19	0.46	0
6	0.45	0
9	0.38	1
16	0.24	1
1	0.23	0
18	0.23	0
5	0.13	0
7	0.12	1
15	0.12	0
10	0.11	0
17	0.1	0
12	0.09	0
20	0.08	1
14	0.07	0
8	0.03	0
11	0.03	0
3	0.01	0

- 例如置信度为0.7时，只有4号框和2号框被保留了下来，查全率为2/6，查准率为2/2。
- 置信度为0.3时，保留了六个框，其中三个检测正确，查全率为3/6，查准率为3/6。
- 因此从趋势的角度而言，置信度越高，意味着模型对剩下的检测框越自信，查准率就越高，但意味着会有很多漏检（查全率低）
- 置信度越低，则剩下的检测框越多，漏检就越低，但产准率也会降低。

检测任务评价指标mAP



- mAP是不同查全率下，最高查准率的平均。

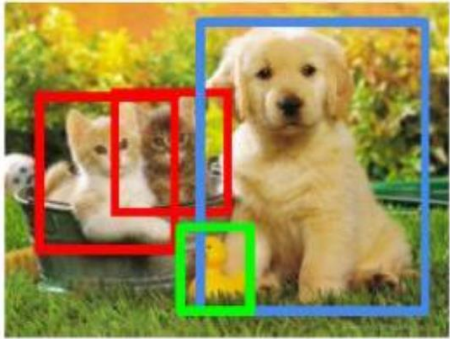
id	score	gt_label	top-N	Precision	Recall(r)	Max Precision for Any Recall $r' \geq r$
4	0.91	1	1	1/1	1/6	1
2	0.76	1	2	2/2	2/6	1
13	0.65	0	3	2/3		
19	0.46	0	4	2/4		
6	0.45	0	5	2/5		
9	0.38	1	6	3/6	3/6	3/6
16	0.24	1	7	4/7	4/6	4/7
1	0.23	0	8	4/8		
18	0.23	0	9	4/9		
5	0.13	0	10	4/10		
7	0.12	1	11	5/11	5/6	5/11
15	0.12	0	12	5/12		
10	0.11	0	13	5/13		
17	0.1	0	14	5/14		
12	0.09	0	15	5/15		
20	0.08	1	16	6/16	6/6	6/16
14	0.07	0	17	6/17		
8	0.03	0	18	6/18		
11	0.03	0	19	6/19		
3	0.01	0	20	6/20		

$$mAP = (1 + 1 + 3/6 + 4/7 + 5/11 + 6/16) / 6$$

深度学习神经网络评价标准



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	分类	定位+分类	物体检测
图示	 CAT	 CAT	 CAT, DOG, DUCK
输入	single and big object	single and big object	multi and small object
输出	label	label & bounding box	multi label & bounding box
评价	precision(top1/top5)	IoU(交并比)	mAP(Mean Average Precision)

/ Network结构的发展与讨论

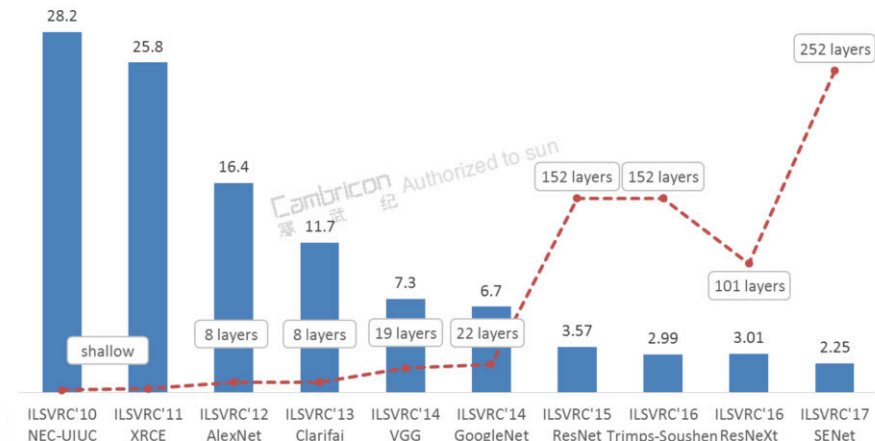


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- 分类任务
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 - VGG-net
 - ResNet
- 检测任务
 - Two-stage model
 - One-stage model
- 序列模型
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 - Transformers

基于CNN的图像分类算法

- 对卷积神经网络的研究可追溯至日本学者福岛邦彦提出的neocognition模型。在其1979 和1980年发表的论文中，福岛参照生物的视觉皮层（visual cortex）设计了以“neocognition”命名的神经网络。
- AlexNet使用卷积神经网络解决图像分类问题，在ILSVRC2012中取得获胜并大大提高了state-of-art的准确率。自此卷积神经网络在图像分类领域获得快速发展。



/ImageNet DataSet

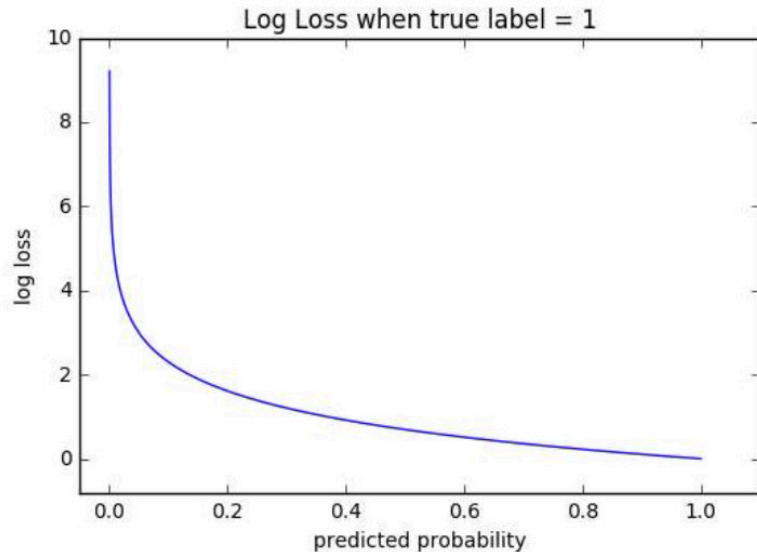
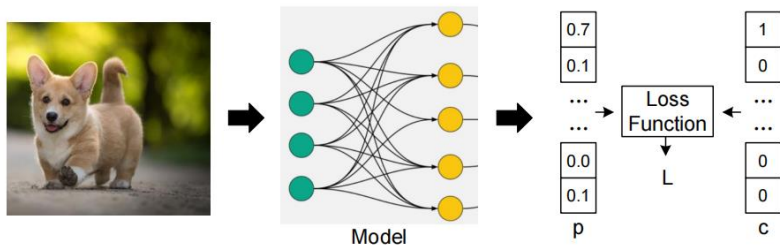
- ImageNet LSVRC-2010 with 1.2 million images
- Top-1 and top-5 error rate:
 - The fraction of images for which the correct label is not among the 1/5 labels considered most probable by the model.



/ AlexNet的成本函数



- Minimize the cross-entropy loss function
- Measures the performance of a classification model whose output is a probability value between 0 and 1
- $CF = -\frac{1}{N} (\sum_{i=1}^N y_i \cdot \log(\hat{y}_i))$



/ AlexNet的优化方法



- Stochastic Gradient Descent

- Batch size = 128
- Update rule:

$$\begin{aligned} v_{i+1} &:= \overset{\text{Momentum}}{0.9 \cdot v_i} - \overset{\text{Weight Decay}}{0.0005 \cdot \epsilon \cdot w_i} - \epsilon \cdot \left\langle \frac{\partial L}{\partial w} \Big|_{w_i} \right\rangle_{D_i} \\ w_{i+1} &:= w_i + v_{i+1} \end{aligned}$$

Learning rate

Gradient of cost function

w: 参数

v: 参数变化量

AlexNet的网络结构



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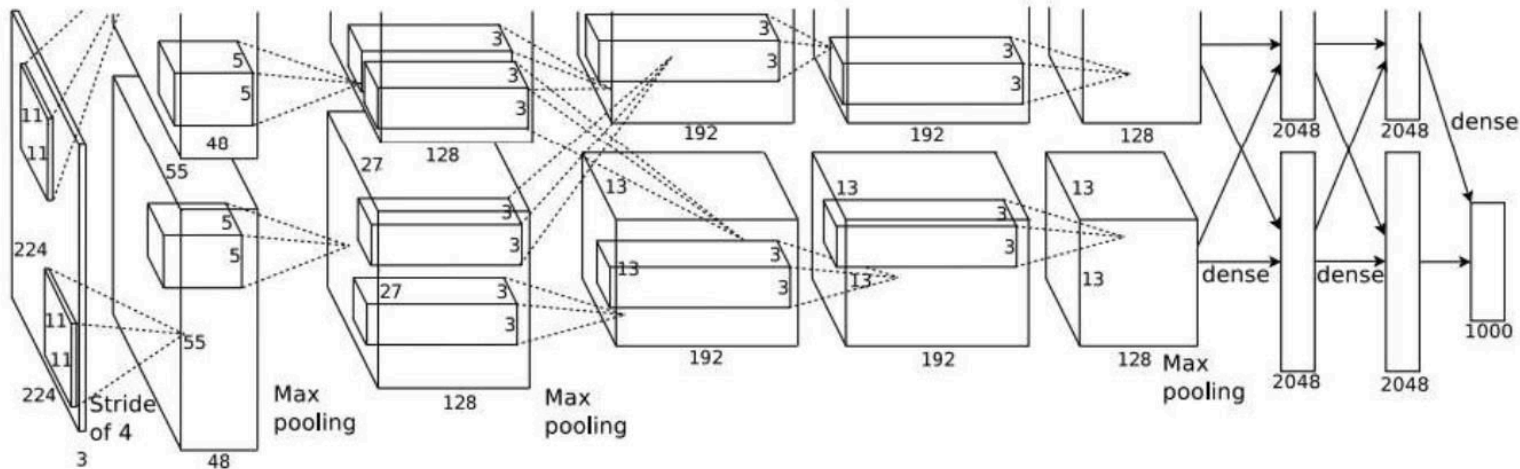
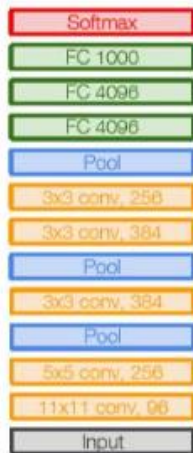


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

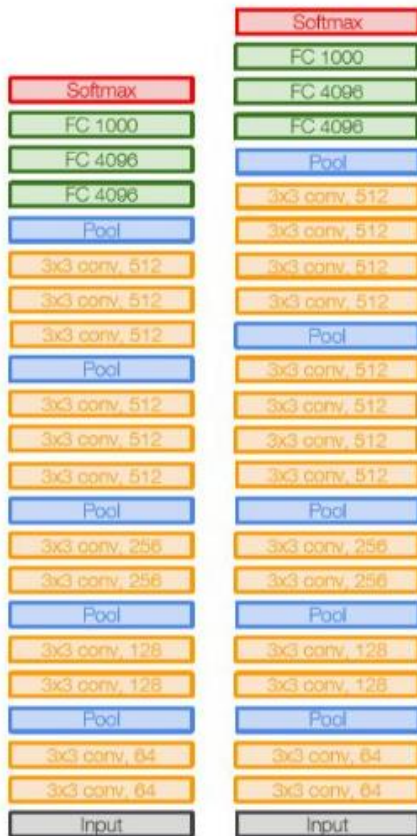
- AlexNet的贡献
 - 使用多个卷积层，有效提取图像特征
 - ReLU帮助提高训练速度
 - Dropout、数据增强扩大训练集，防止过拟合

使用更多卷积层是否能进一步提升效果？





AlexNet



VGG16

VGG19

AlexNet和VGG有

什么不同的地方？

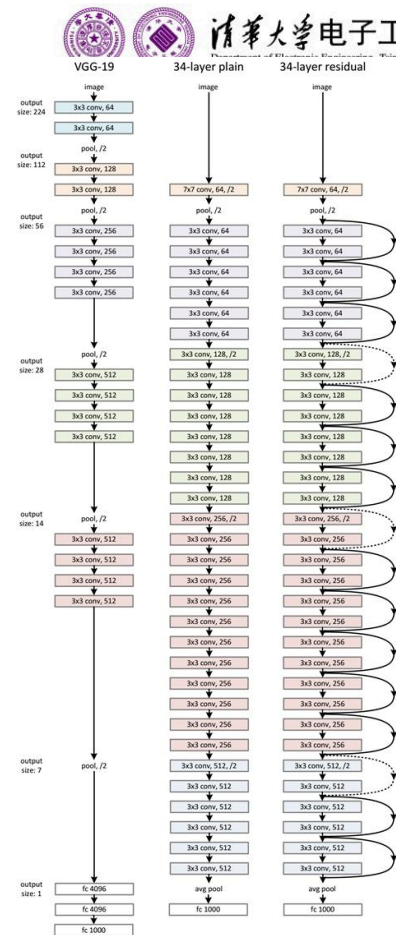
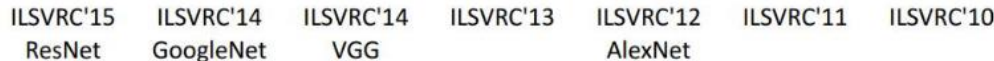
不同深度VGG的结构

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

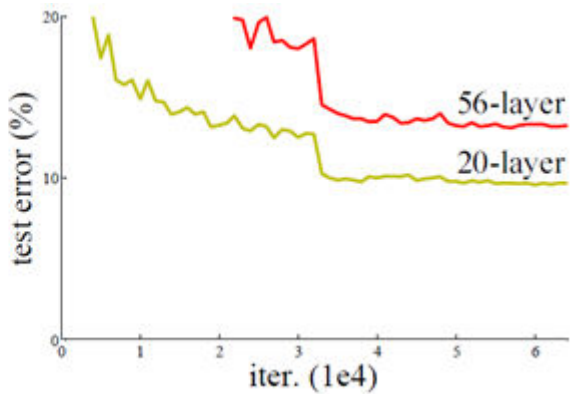
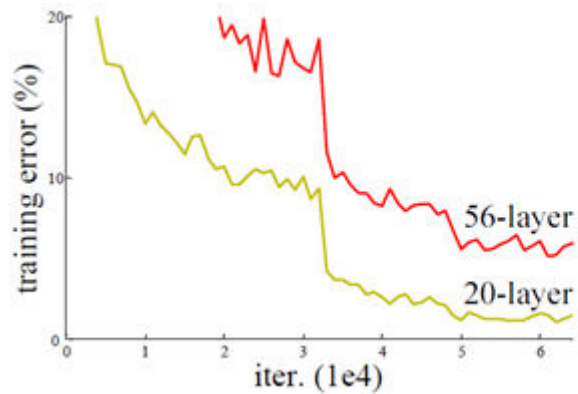
VGG的贡献

- 用3*3的卷积核取代AlexNet中的大卷积核
- 参数预初始化策略
- 为什么3*3更好？
 - 提升收敛速度
 - 参数量更少
 - 可以构建更深的网络，有更多的非线性变换，更强的表征能力

152 layers

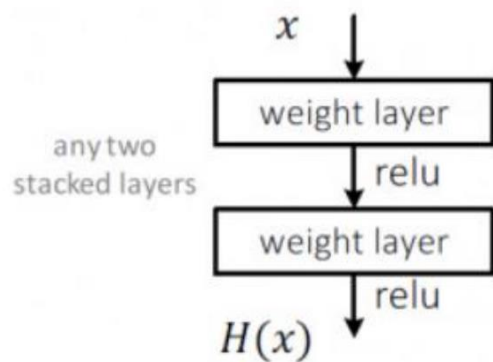


- 问题：为什么前人训练不成这么深的网络呢？
- ResNet的作者分别用20层和56层的网络在cifar10上做了训练和测试，发现56层的网络不仅测试错误率高，而且训练集上错误率也很高。

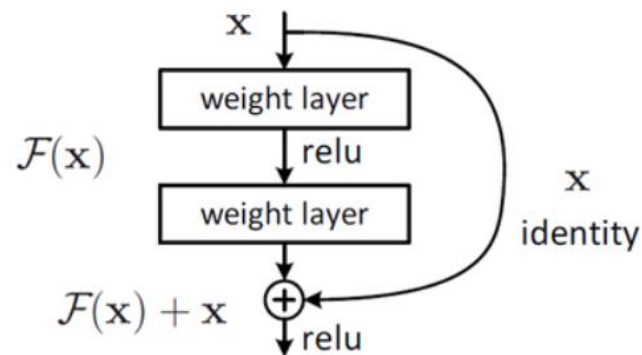


- 过拟合？
 - 过拟合的话，在训练集上精度应该更高才对，事实相反。
- 深层网络的退化问题。
 - 现有优化技术很难优化深层网络，致使深层网络只是不断复制浅层特征。

Plain-net (VGG-style network)



ResNet



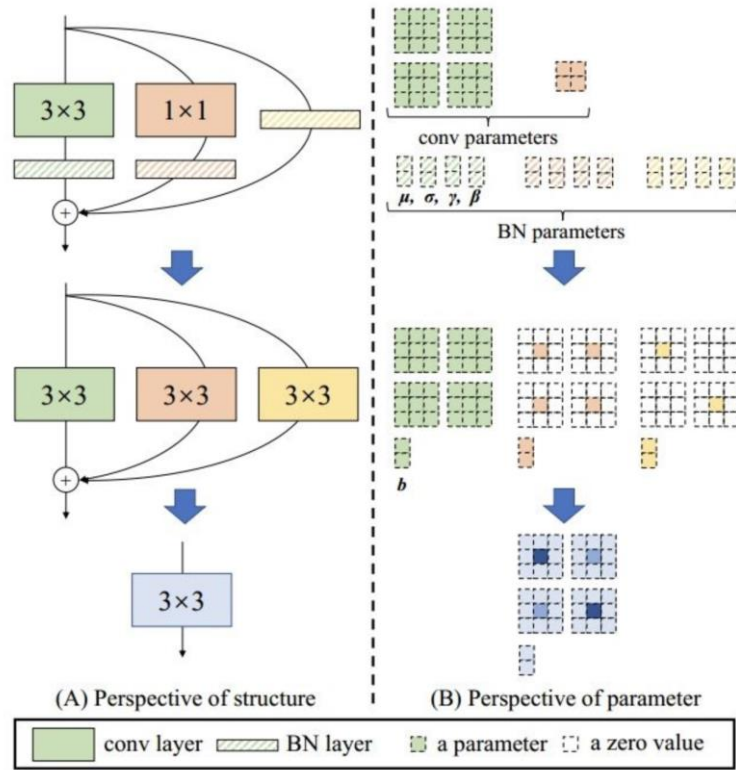
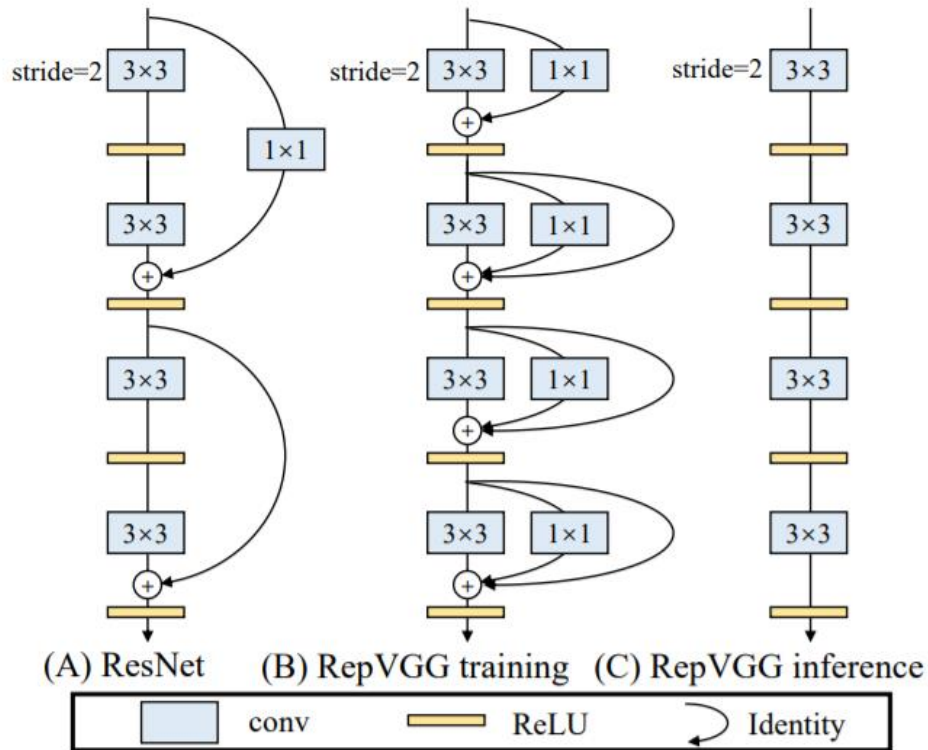
RepVGG



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Make VGG-style Convnets great again

<https://zhuanlan.zhihu.com/p/344324470>



/ Network结构的发展与讨论



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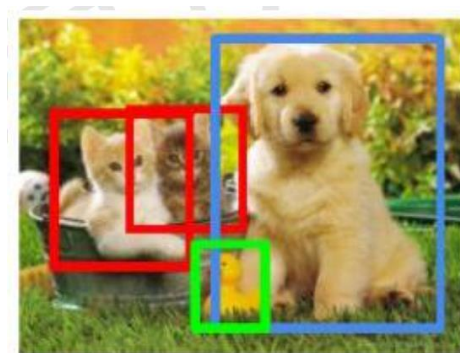
深度学习神经网络：目标检测



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基于CNN的目标检测算法

- two-stage算法：先生成边界框，再做分类。代表工作：R-CNN系列
- one-stage算法：同时输出定位和分类。代表工作：Yolo系列（you only look once）



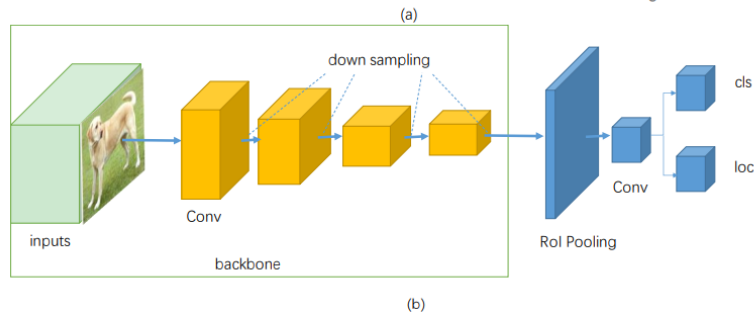
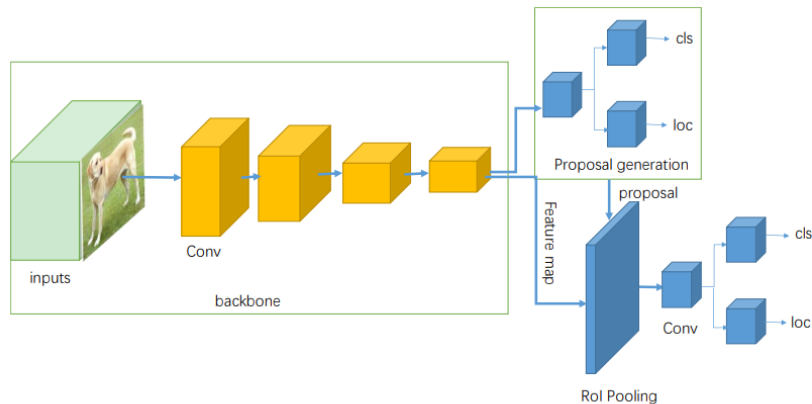
CAT, DOG, DUCK

Two-stage v.s. One-stage Algorithms



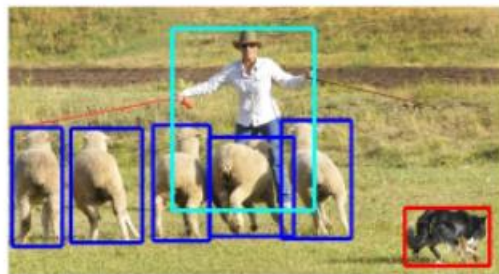
Two stage: classification after generating binding boxes.

One stage: generating binding boxes and types simultaneously.

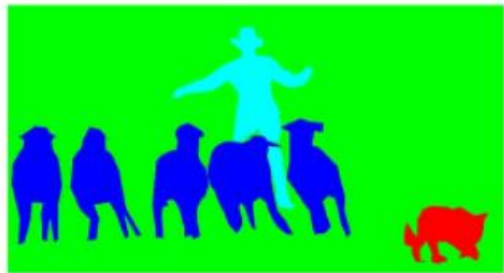




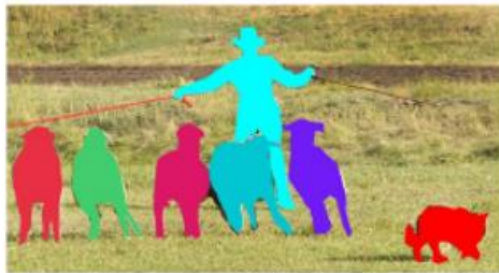
(a) Image classification



(b) Object detection



(c) Semantic segmentation



(d) Instance segmentation

- COCO 数据集包含分类、检测、分割等多项视觉任务。其具有超过30万张图像，和超过200万的标签。
- 其被用于最多的，还是目标检测任务。

/R-CNN family



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R-CNN [Girshick *et.al.*, CVPR 2014]

mAP: 53.3%

latency: 50s

feature: region proposal, CNN feature extraction, SVM classification

Fast R-CNN [Girshick *et.al.*, ICCV 2015]

mAP: 65.7%

latency: 2s

feature: ROI Pooling, softmax classification

Faster R-CNN [Ren *et.al.*, NIPS 2015]

mAP: 67.0%

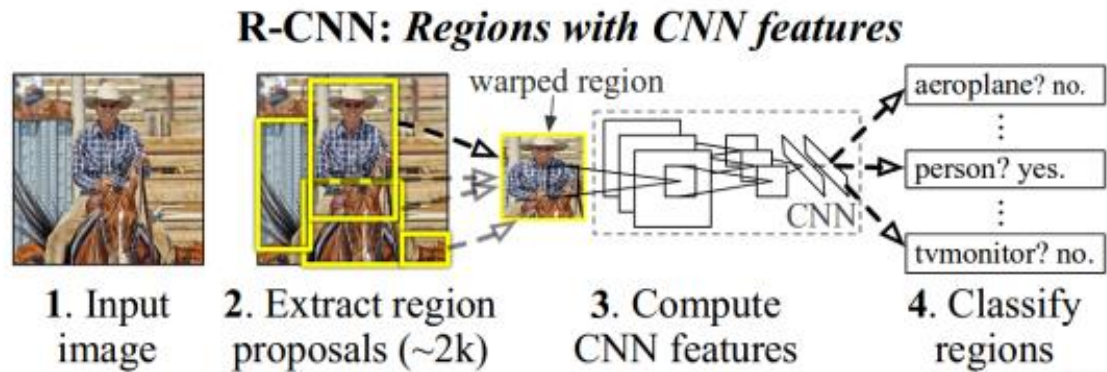
latency: 0.2s

feature: Region Proposal Network (RPN)

Two-stage 检测模型 R-CNN



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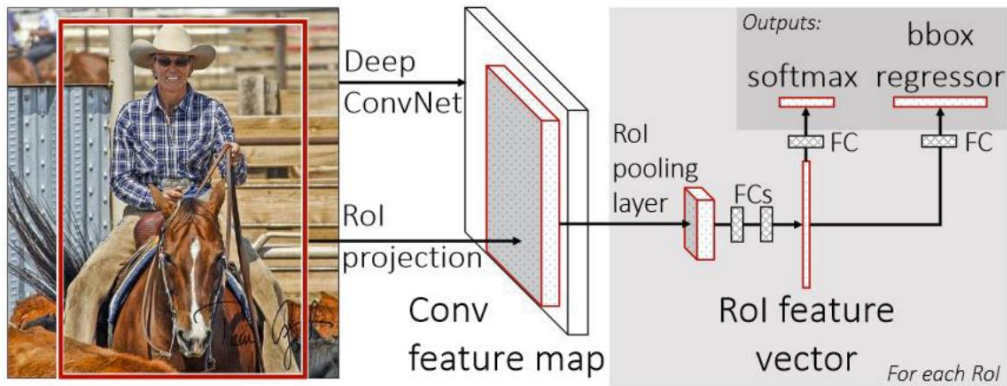
- 输入图像
- 提取候选框
- 每一个候选框单独提取特征
- 进行分类

Two-stage即为两步走，先画框，再分类

Two-stage 检测模型 Fast R-CNN



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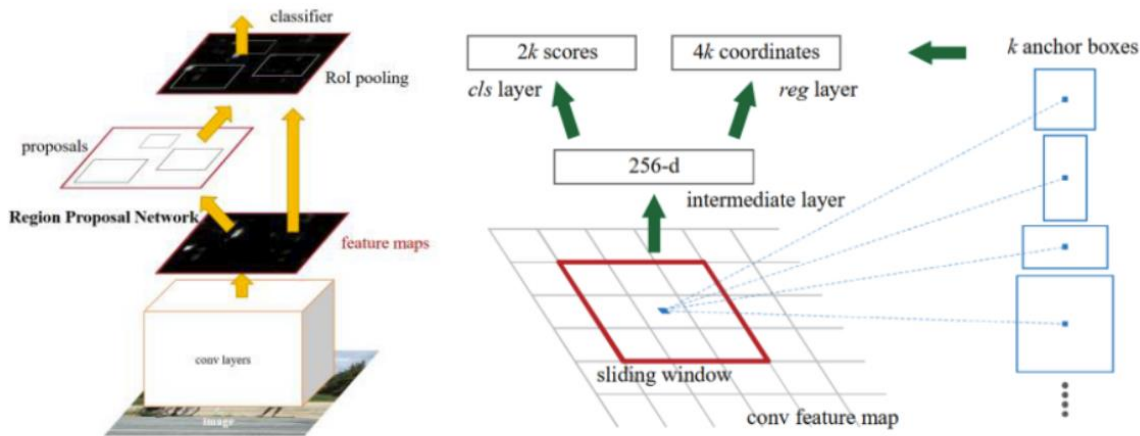
- 输入图像
- 一次特征提取
- 提取候选框（选择性搜索算法）
- 进行分类

Two-stage即为两步走，先画框，再分类

Two-stage 检测模型 Faster R-CNN



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- 输入图像
- 一次特征提取
- 提取候选框（神经网络提取RPN）
- 进行分类

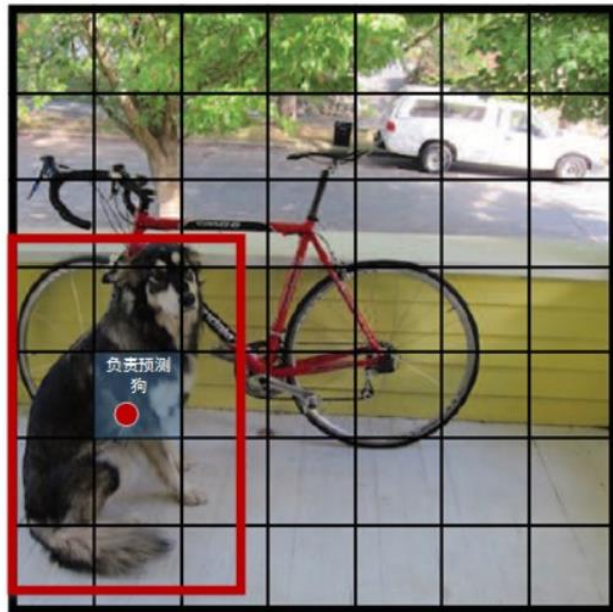
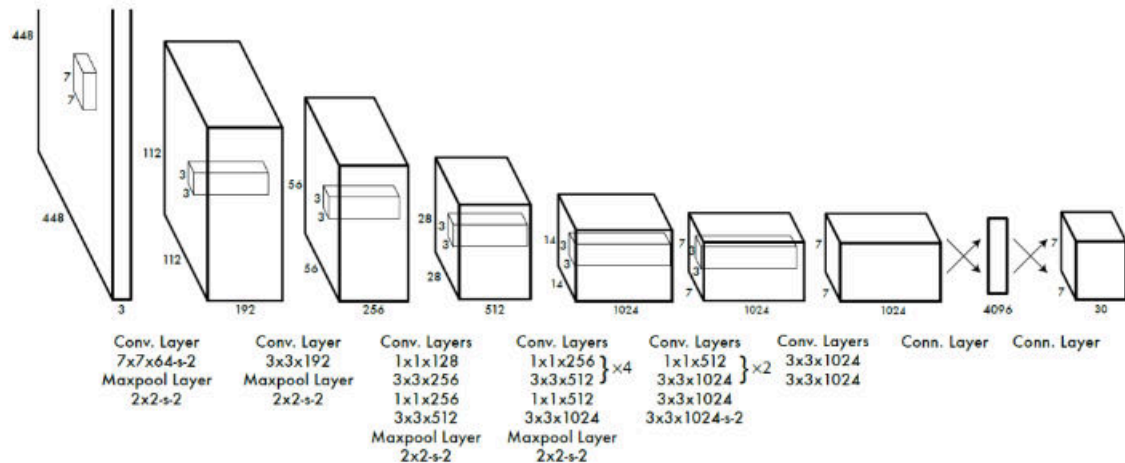
Two-stage即为两步走，先画框，再分类

One-stage 检测模型: YOLO



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网络结构



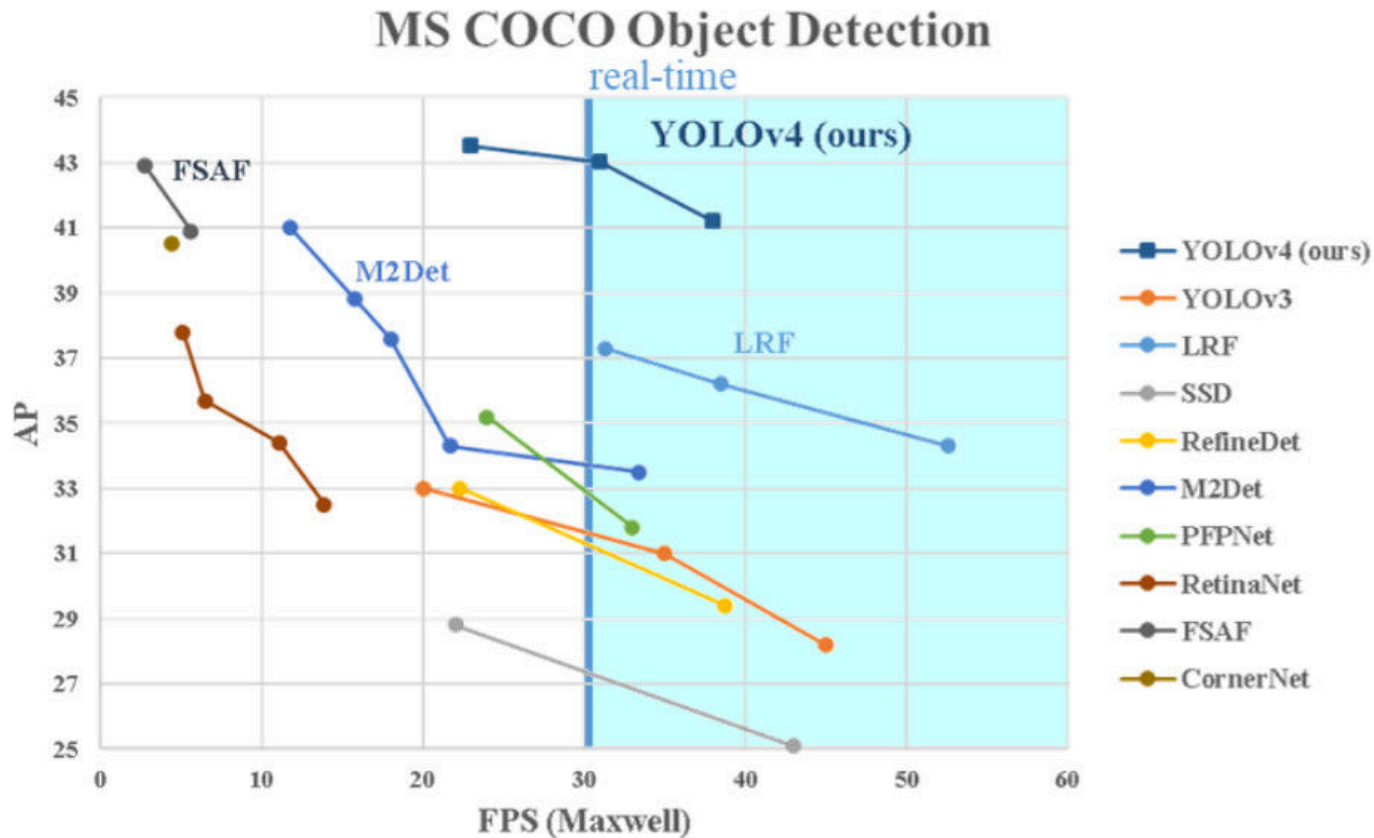
经过池化层后，输入图像的尺寸会减小。经过更多池化层后，相当于当前像素是由之前的一个图像块生成的。

/ One-stage 检测模型: YOLO



- YOLO对于每个像素，都会最终输出一个对应的特征向量，包含：
 - 该像素是物体中心的置信度（二分类）
 - 该像素坐标偏离物体中心的长度 Δx 和 Δy （回归）
 - 该像素所代表物体的分类，及置信度（分类）
 - 该像素所代表物体的长宽，yolo会预先设定一些基础框（anchor），输出的则是相对于这些基础框的形变 Δh 和 Δw （回归）
- YOLO先后经历了很多代发展，已经成为使用最广泛的目标检测算法

FPS: 每秒
处理的帧
数



/NMS (Non-Maximum Suppression)

Bounding boxes for one instance may overlap.

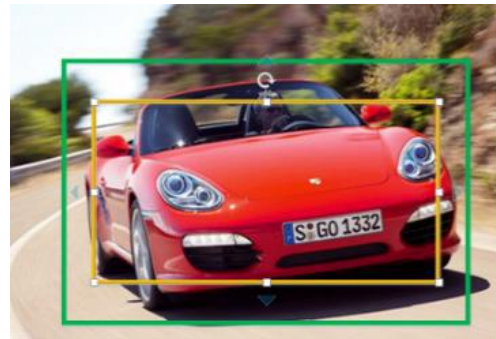
Method: For each type, use NMS to eliminate redundant bounding boxes (greedy approach).

Workflow:

1. Sort candidate bounding boxes by classification confidence.
2. Adding the boxes ***b*** with most confidence to output list, and delete it from the candidate boxes.
3. Calculate IoU between ***b*** and other boxes ***bi***. If $>$ threshold, delete ***bi***.
4. Repeat until no candidate bounding boxes.



NMS (type-dependent)



/ Network结构的发展与讨论



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Serial Model



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Applications

- Natural Language Processing
 - Language Modeling
 - Speech Recognition
 - Machine Translation
 - Conversation Modeling
 - Question Answering
- Computer Vision
 - Object Recognition
 - Image Generation
 - Video Analysis
- Multimodal (CV+NLP)
 - Image Captioning
 - Video Captioning
 - Visual Question Answering
- Turing Machines
- Robotics
- Other

Serial Model: to process Speech, text, video, audio, etc.

Feature:

1. The data input is in the time sequence.
2. There is a correlation between the data before and after.
So the model should have the ability to "store" information.

美国总统布什昨天在白宫与以色列总理沙龙就中东局势举行了一个小时的会谈。

US President George W. Bush held an hour-long meeting with Israeli Prime Minister Ariel Sharon on the situation in the Middle East yesterday at the White House.

Translate



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."



"young girl in pink shirt is swinging on swing."



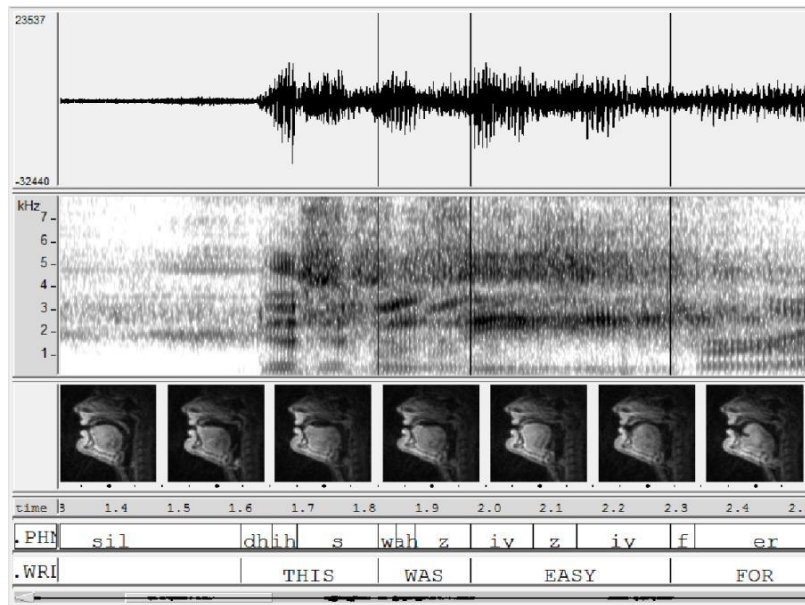
"man in blue wetsuit is surfing on wave."

Image Captioning

Speech dataset: TIMIT dataset

It consists of recordings of 630 speakers of 8 dialects of American English each reading 10 phonetically-rich sentences.

It also comes with the word and phone-level transcriptions of the speech.



Video dataset: DAVIS

The Densely Annotation Video Segmentation dataset (**DAVIS**) is a high quality and high resolution densely annotated video segmentation dataset under two resolutions, 480p and 1080p.

There are 50 video sequences with 3455 densely annotated frames in pixel level. 30 videos with 2079 frames are for training and 20 videos with 1376 frames are for validation.



/NLP dataset: GLUE

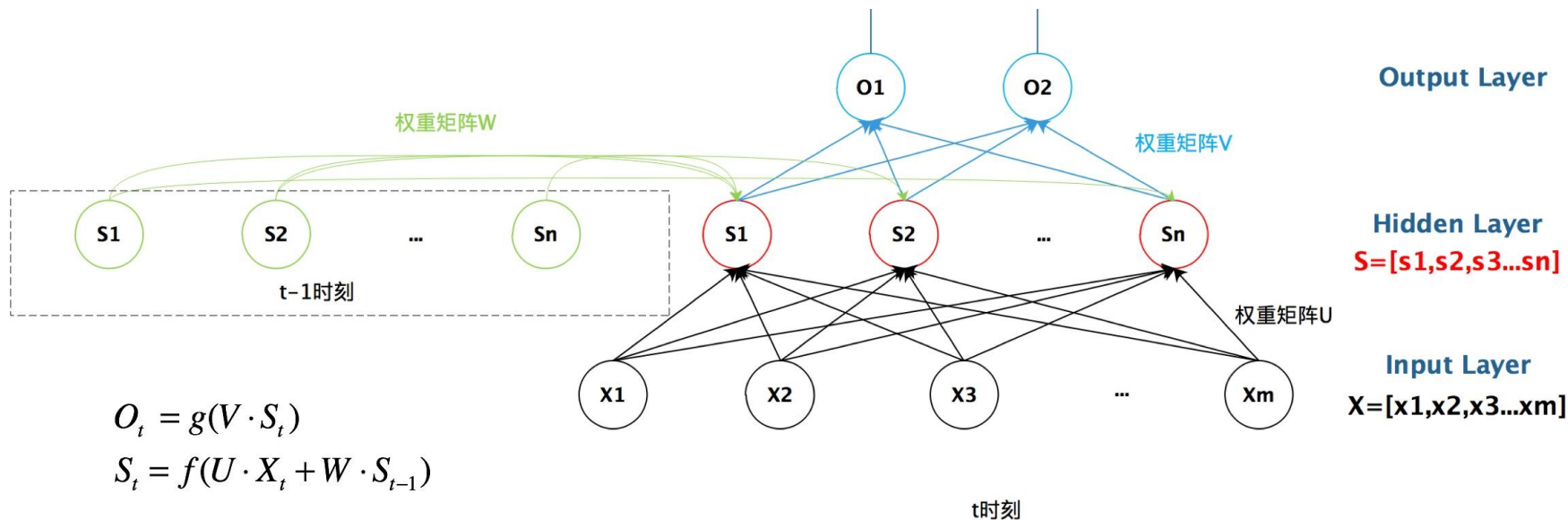


- General Language Understanding Evaluation (**GLUE**) benchmark: Standard split of data to train, validation, test, where labels for the test set is only held in the server.
- Sentence pair tasks
 - **MNLI**, Multi-Genre Natural Language Inference
 - **QQP**, Quora Question Pairs
 - **QNLI**, Question Natural Language Inference
 - **STS-B** The Semantic Textual Similarity Benchmark
 - **MRPC** Microsoft Research Paraphrase Corpus
 - **RTE** Recognizing Textual Entailment
 - **WNLI** Winograd NLI is a small natural language inference dataset
- Single sentence classification
 - **SST-2** The Stanford Sentiment Treebank
 - **CoLA** The Corpus of Linguistic Acceptability

/ RNN: Recurrent Neural Network



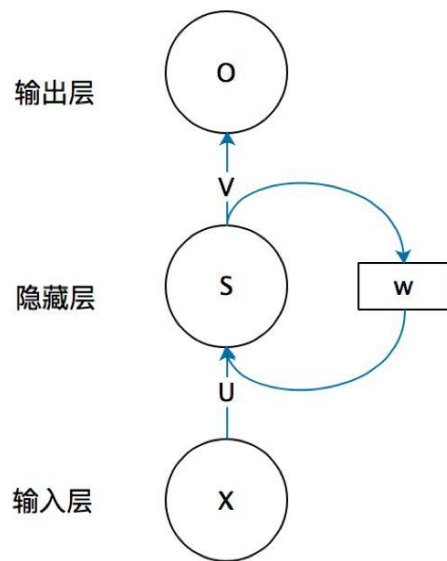
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/ RNN: Recurrent Neural Network

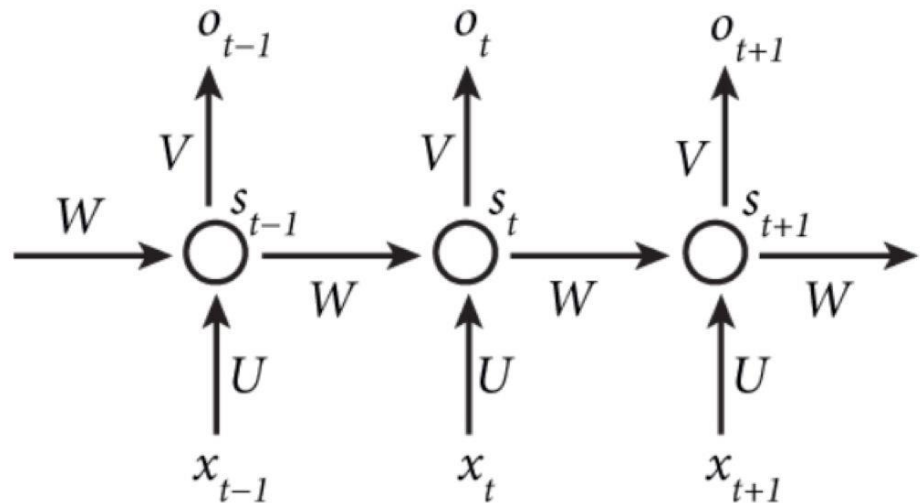


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循环层

按照时间线展开

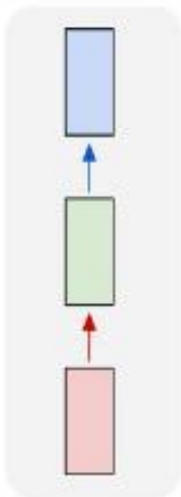


Multiple RNN input output structure



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one to one



one to many

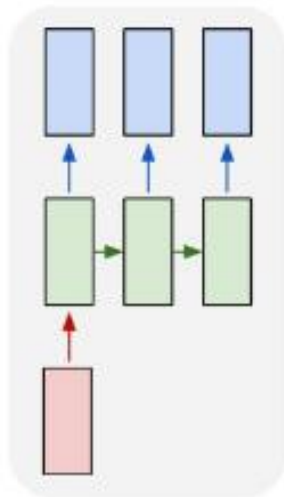
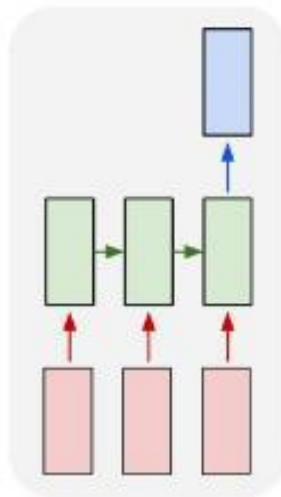


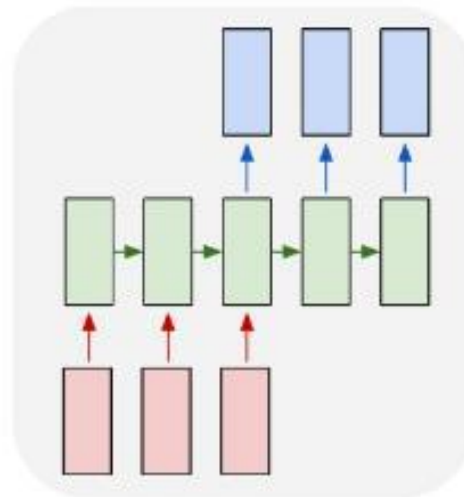
Image
captioning

many to one



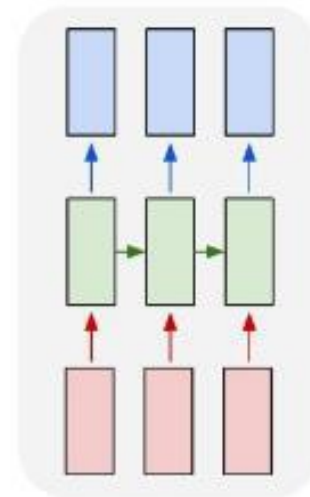
sentiment
analysis

many to many



machine
translation

many to many



video
classification

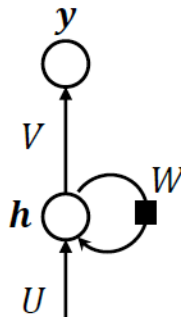
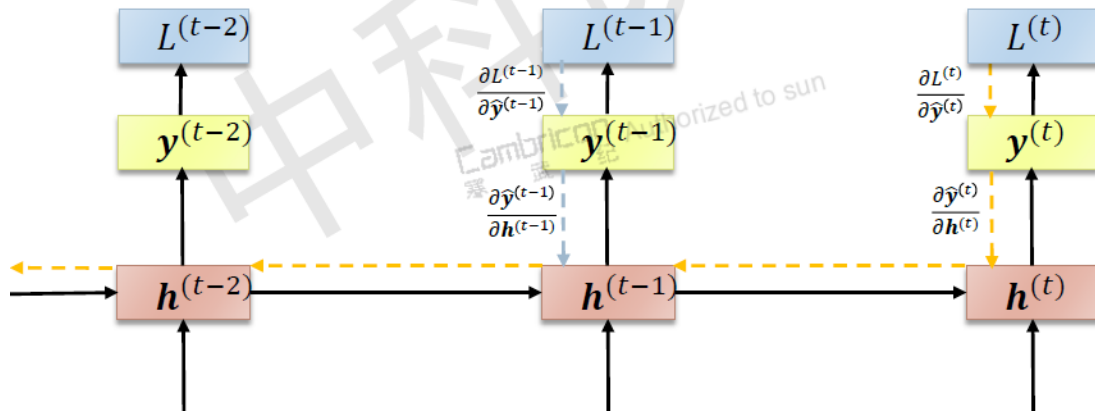
RNN cost function



某个时刻的损失函数为: $L^{(t)} = -y^{(t)} \ln \hat{y}^{(t)}$

整个序列的损失函数为: $L = \sum_{t=1}^{\tau} L^{(t)} = - \sum_{t=1}^{\tau} y^{(t)} \ln \hat{y}^{(t)}$

损失函数对W的偏导为: $\frac{\partial L}{\partial W} = \sum_{t=1}^{\tau} \frac{\partial L^{(t)}}{\partial W} = \sum_{t=1}^{\tau} \sum_{k=1}^t \frac{\partial J^{(t)}}{\partial \hat{y}^{(t)}} \frac{\partial \hat{y}^{(t)}}{\partial \mathbf{h}^{(t)}} \frac{\partial \mathbf{h}^{(t)}}{\partial \mathbf{h}^{(k)}} \frac{\partial \mathbf{h}^{(k)}}{\partial W}$



Gradient vanishing/exploding



▸ 梯度爆炸/梯度消失

根据推导可知序列损失函数对U和W的梯度为：

$$\frac{\partial L}{\partial W} = \sum_t \sum_{k=1}^{k=t} \frac{\partial L^{(t)}}{\partial \hat{\mathbf{y}}^{(t)}} \frac{\partial \hat{\mathbf{y}}^{(t)}}{\partial \mathbf{o}^{(t)}} \frac{\partial \mathbf{o}^{(t)}}{\partial \mathbf{h}^{(t)}} \left(\prod_{i=k+1}^t W^\top \text{diag} \left(1 - (\mathbf{h}^{(i)})^2 \right) \right) \frac{\partial \mathbf{h}^{(k)}}{\partial W}$$

$$\frac{\partial L}{\partial U} = \sum_t \sum_{k=1}^{k=t} \frac{\partial L^{(t)}}{\partial \hat{\mathbf{y}}^{(t)}} \frac{\partial \hat{\mathbf{y}}^{(t)}}{\partial \mathbf{o}^{(t)}} \frac{\partial \mathbf{o}^{(t)}}{\partial \mathbf{h}^{(t)}} \left(\prod_{i=k+1}^t W^\top \text{diag} \left(1 - (\mathbf{h}^{(i)})^2 \right) \right) \frac{\partial \mathbf{h}^{(k)}}{\partial U}$$

$$\text{令 } \gamma = \left\| \prod_{i=k+1}^t W^\top \text{diag} \left(1 - (\mathbf{h}^{(i)})^2 \right) \right\|_2$$

$$\text{当 } t \gg k \text{ 时, } \gamma \begin{cases} \rightarrow \infty, & \frac{\partial L}{\partial U} \rightarrow \infty, \frac{\partial L}{\partial W} \rightarrow \infty \text{ 梯度爆炸} \\ \rightarrow 0, & \frac{\partial L}{\partial U} \rightarrow 0, \frac{\partial L}{\partial W} \rightarrow 0 \text{ 梯度消失} \end{cases}$$

由于梯度爆炸或消失的存在, 循环神经网络实际上只能学习到短期的依赖关系。

/ Gradient vanishing/exploding



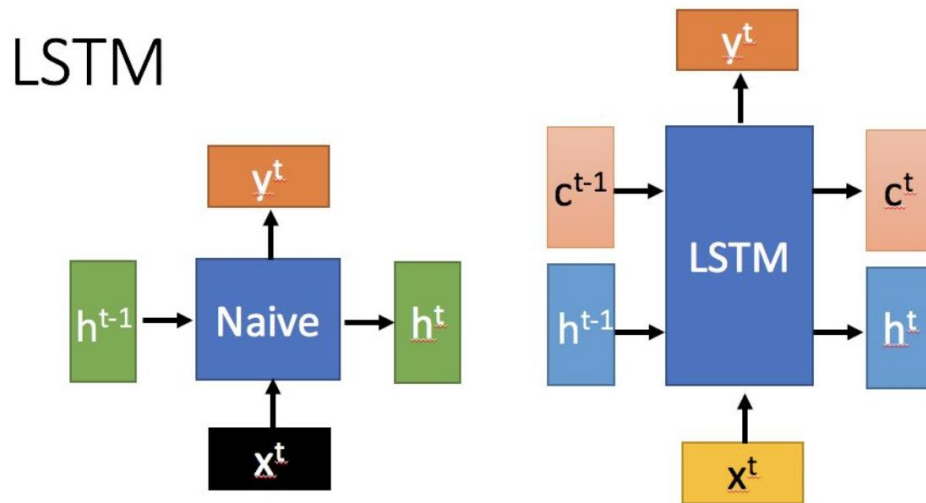
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- ▶ 由于梯度消失，循环神经网络无法处理长期依赖关系；
- ▶ 比如，考虑一个语言模型，试图根据之前单词预测下一个。
- ▶ 如果要预测“The clouds are in the sky”中最后一个单词，不需要更多的上下文即可知道下一个单词会是“sky”。在这种情况下，相关信息与预测位置的间隔比较小，RNNs可以学会使用之前的信息。
- ▶ 考虑试图预测“I grew up in Italy... I speak fluent Italian.”中最后一个，则需要用到包含“Italy”的上下文，从前面的信息推断后面的单词。相关信息与预测位置的间隔可能会很大。随着这种间隔的拉长，RNNs就会无法学习连接信息。

如何记住长期依赖？



/LSTM: solving the gradient



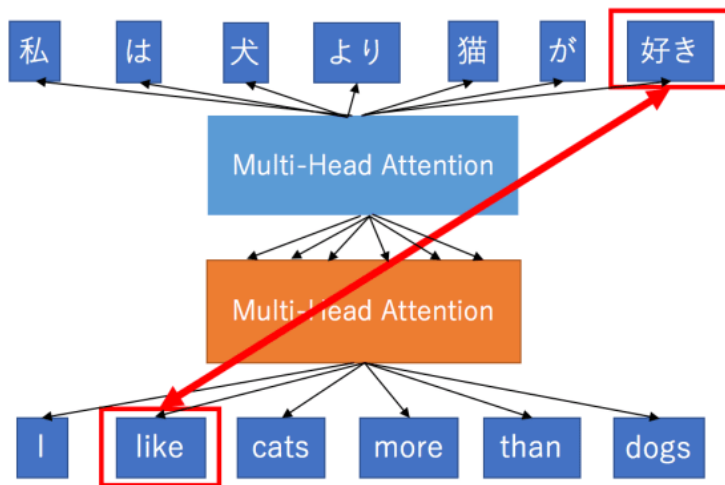
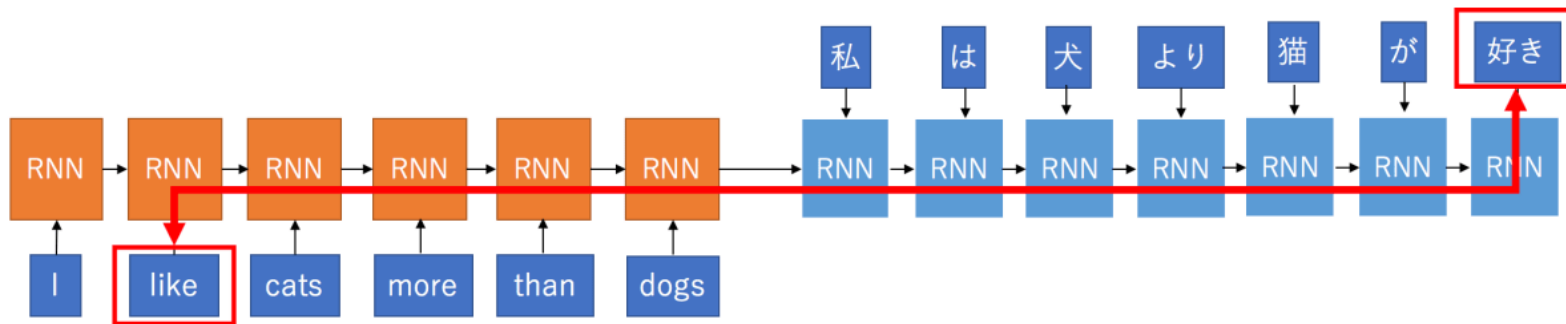
c change slowly $\Rightarrow c^t$ is c^{t-1} added by something

h change faster $\Rightarrow h^t$ and h^{t-1} can be very different

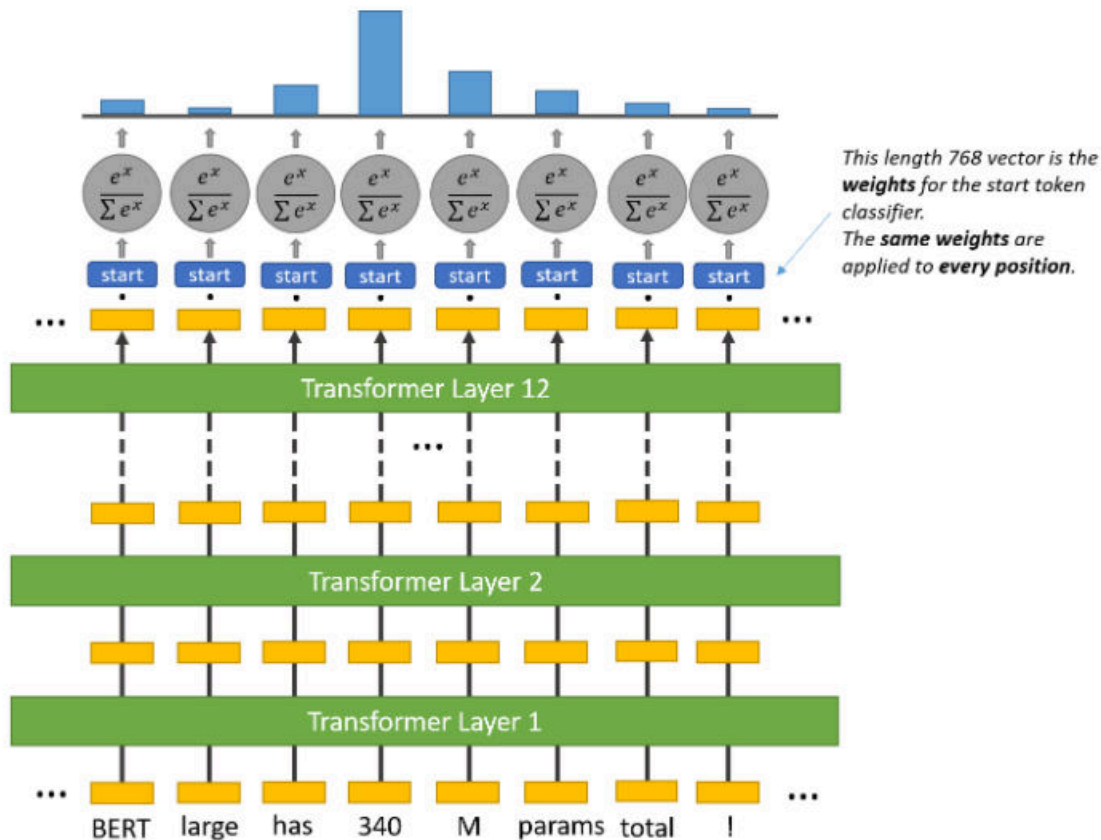
Maximum Path Length: RNN v.s. Transformer



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/ Bert: network architecture figure.



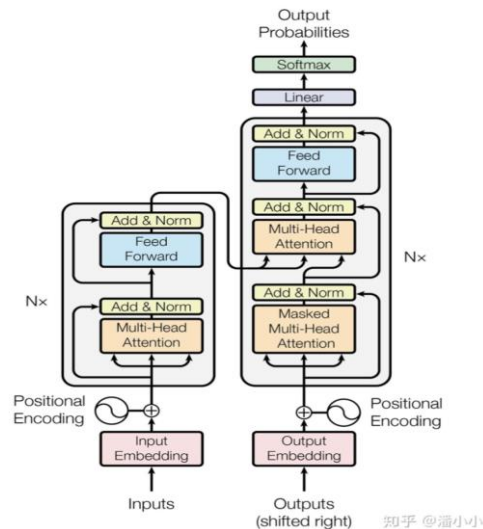
Transformer: Basic build block for NLP networks.



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Transformer encoder

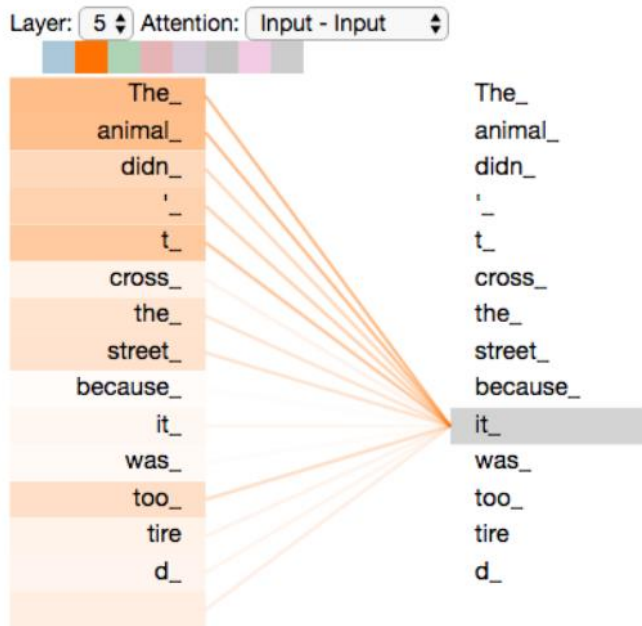
- Multi-headed self attention
 - Models context
- Feed-forward layers
 - Computes non-linear hierarchical features
- Layer norm and residuals
 - Makes training deep networks healthy
- Positional embeddings
 - Allows model to learn relative positioning



编码器

解码器

/ Self attention

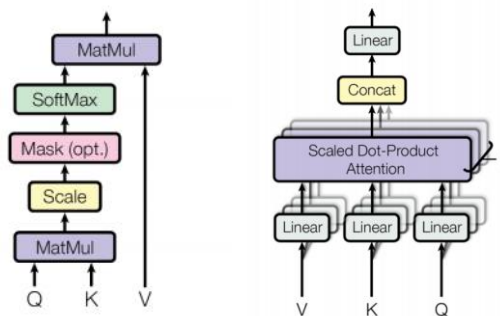


"The animal didn't cross the street because it was too tired"

"The animal didn't cross the street because it was too wide"

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Scaled Dot-Product Attention



Acknowledgement to Figure from <http://jalamar.github.io/illustrated-bert/>

Equation and Figure in (Vaswani *et al.*, 2017)

Bert: Bidirectional Encoder Representations from Transformers

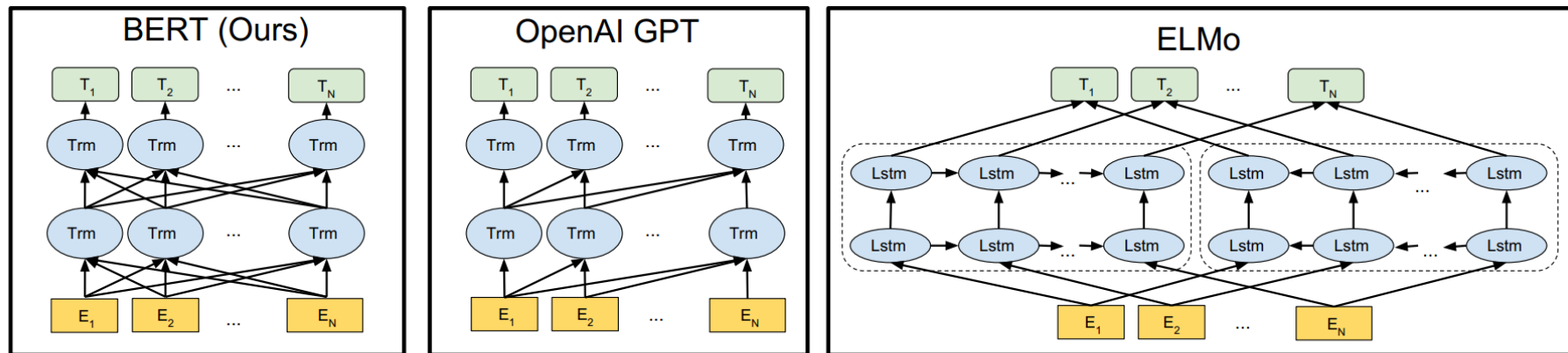
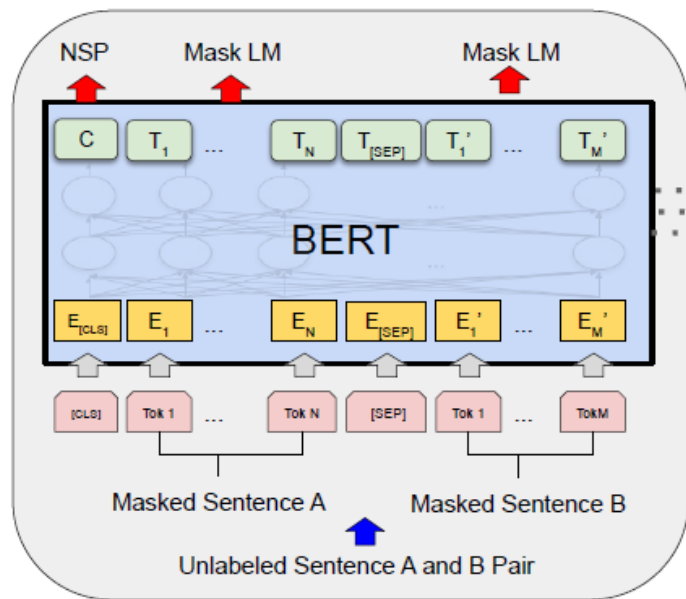
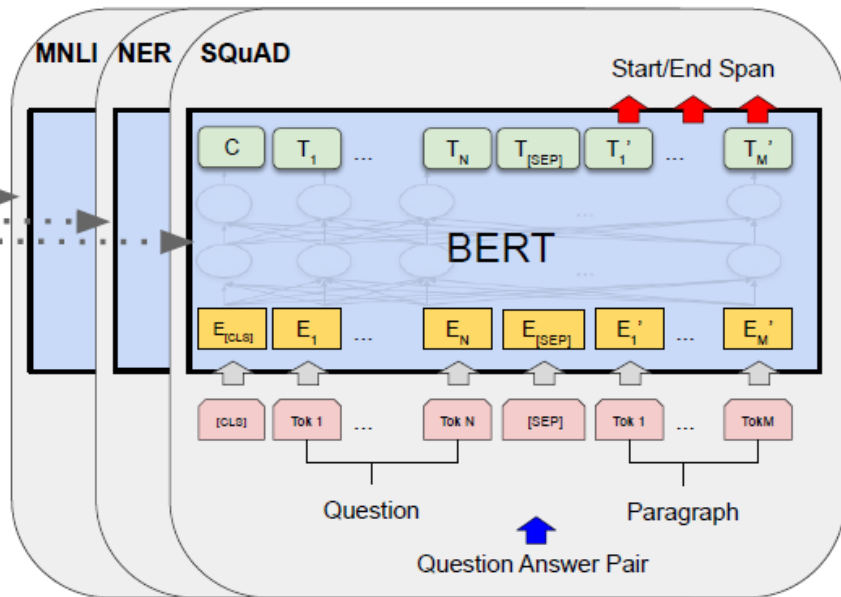


Figure 1: Differences in pre-training model architectures. BERT uses a bidirectional Transformer. OpenAI GPT uses a left-to-right Transformer. ELMo uses the concatenation of independently trained left-to-right and right-to-left LSTM to generate features for downstream tasks. Among three, only BERT representations are jointly conditioned on both left and right context in all layers.

/ Bert: pretrained model and tuning to various tasks: making it affordable.



Pre-training



Fine-Tuning



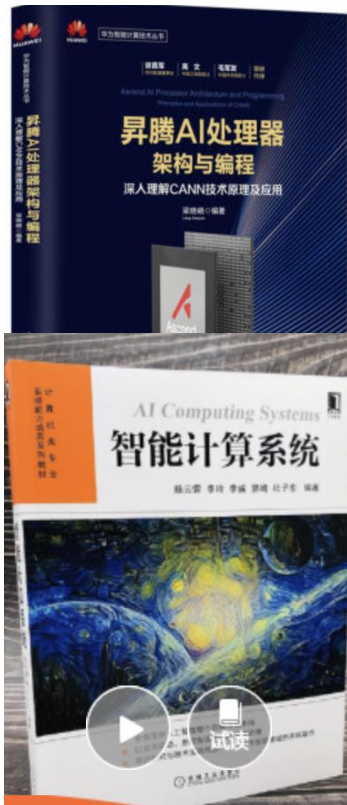
- **Classification App**
 - **Dataset: ImageNet**
 - **AlexNet/VGG/ResNet**
- **Detection App**
 - **Dataset: COCO**
 - **Two-stage: R-CNN, SSD**
 - **One-stage: YOLO**
- **Serial App**
 - **Dataset**
 - **RNN**
 - **LSTM**
 - **Transformers**

布置作业



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- 1) Reading for this week
 - ResNet/Yolo Paper
 - Submit your review before next course.
- 2) Book Reading
 - 中文教材第一章和第六章
- 3) Lab 1 will be posted after next course
 - Quantization of CNN





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寒武纪 智能计算系统 课程

三位助教帮助完成课件整理